Krill oil supplementation with and without resistance training lowers the burden of the metabolic syndrome but does not affect glucose tolerance in sedentary adults: a randomized controlled trial

Håvard Hamarsland, Magnus Husøy, Håvard Nygaard, Solveig Vuilliomenet, Jørn K. Thorgersen, Even H. Rosenlund, Emilie S. Larsen, Øistein Hovde, Daniel Hammarström, Stian Ellefsen

21 October 2026

Håvard Hamarsland[1†], Magnus Husøy[1,2†], Håvard Nygaard[1], Solveig Vuilliomenet[1], Jørn K. Thorgersen[1], Even H. Rosenlund[1], Emilie S. Larsen[1], Øistein Hovde[3], Daniel Hammarström[1], Stian Ellefsen[1,3*]

[1] Section for Health and Exercise Physiology, University of Inland Norway, Lillehammer, Norway
[2] Laboratory of Medical Biochemistry, Innlandet Hospital Trust, Lillehammer, Norway
[3] Innlandet Hospital Trust, Brumunddal, Norway

[†] These authors contributed equally to this work.
[*] Corresponding author: Stian Ellefsen, stian.ellefsen@inn.no

Contents

1 Introduction . . . 5
1.1 Project organisation . . . 5
1.1.1 R scripts . . . 5
1.1.2 Stan models . . . 5
1.2 Packages and computational environment . . . 5

2 Analytic approach and data description . . . 7
2.1 Data analysis . . . 7
2.2 Description of the data set and participants . . . 7

3 Effects of N-3 supplementation of serum fatty acids . . . 11
3.0.1 Model expansion: Heterogeneous variance . . . 24
3.1 Inference . . . 26

4 Effects of krill oil supplementation on indications of metabolic syndrome - Introduction 31
4.1 Effects of N-3 supplementation on indicies of metabolic syndrome . . . 31
4.1.1 Synthetic data . . . 39

5 Effects of N-3 supplementation on indications of metabolic syndrome - Models 47
5.0.1 Prior predictive checks, preliminary models . . . 49

Bibliography . . . 59

1. Introduction

This document includes supplementary data and computational details related to the paper "Krill oil supplementation with and without resistance training lowers the burden of the metabolic syndrome but does not affect glucose tolerance in sedentary adults: a randomized controlled trial".

1.1 Project organisation

This notebook is built from source code available at the github repository trainome/alpha-omega-trial. The data used in analyses is organized as an R data package available in a separate repository trainome/AOData.

1.1.1 R scripts

- `R/fig-opt-fun.R` - Helper functions and settings for data visualisations.
- `R/helpers.R` - Helper functions for modelling.
- `R/packages.R` - Loading required packages.
- `R/data-prep.R` - Prepares data for the Metabolic syndrome analysis.
- `R/sim-fun.R` - Functions related to simulation studies.

1.1.2 Stan models

All Stan model scripts are stored in `/models/`. If model code is executed locally, model executable are also stored here. Model fits are stored in `fits/` but not pushed to github due to their large size.

1.2 Packages and computational environment

Stan (CmdStan) version 2.39.0 was used in compiling this notebook. R packages are listed below together with additional session info.

```
sessionInfo()
```

```
R version 4.6.0 (2026-04-24 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26300)

Matrix products: default
  LAPACK version 3.12.1

locale:
[1] LC_COLLATE=Norwegian Bokmål_Norway.utf8
[2] LC_CTYPE=Norwegian Bokmål_Norway.utf8
[3] LC_MONETARY=Norwegian Bokmål_Norway.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=Norwegian Bokmål_Norway.utf8

time zone: Europe/Oslo
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
```

```
 [1] Gmisc_3.3.0       htmlTable_2.5.0   corrr_0.4.5       loo_2.9.0
 [5] posterior_1.7.0   cmdstanr_0.9.0    ggh4x_0.3.1       gt_1.3.0
 [9] patchwork_1.3.2   ggrepel_0.9.8     ggtext_0.1.2      lubridate_1.9.5
[13] forcats_1.0.1     stringr_1.6.0     dplyr_1.2.1       purrr_1.2.2
[17] readr_2.2.0       tidyr_1.3.2       tibble_3.3.1      ggplot2_4.0.3
[21] tidyverse_2.0.0   AOData_0.0.1.9000

loaded via a namespace (and not attached):
 [1] gtable_0.3.6         tensorA_0.36.2.1     xfun_0.57
 [4] htmlwidgets_1.6.4    processx_3.9.0       lattice_0.22-9
 [7] tzdb_0.5.0           vctrs_0.7.3          tools_4.6.0
[10] generics_0.1.4       parallel_4.6.0       cluster_2.1.8.2
[13] pkgconfig_2.0.3      data.table_1.18.4    checkmate_2.3.4
[16] RColorBrewer_1.1-3   S7_0.2.2             distributional_0.7.0
[19] lifecycle_1.0.5      compiler_4.6.0       farver_2.1.2
[22] forestplot_3.2.0     htmltools_0.5.9      yaml_2.3.12
[25] Formula_1.2-5        pillar_1.11.1        Hmisc_5.2-5
[28] rpart_4.1.27         abind_1.4-8          tidyselect_1.2.1
[31] digest_0.6.39        stringi_1.8.7        fastmap_1.2.0
[34] grid_4.6.0           colorspace_2.1-2     cli_3.6.6
[37] magrittr_2.0.5       base64enc_0.1-6      XML_3.99-0.23
[40] foreign_0.8-91       withr_3.0.2          scales_1.4.0
[43] backports_1.5.1      timechange_0.4.0     rmarkdown_2.31
[46] matrixStats_1.5.0    nnet_7.3-20          otel_0.2.0
[49] gridExtra_2.3        hms_1.1.4            evaluate_1.0.5
[52] knitr_1.51           rlang_1.2.0          gridtext_0.1.6
[55] Rcpp_1.1.1-1.1       glue_1.8.1           xml2_1.5.2
[58] rstudioapi_0.18.0    jsonlite_2.0.0       R6_2.6.1
[61] fs_2.1.0
```

2. Analytic approach and data description

2.1 Data analysis

For inferential statistics, Bayesian models were constructed using the Stan probabilistic programming language (1) and fitted using the `cmdstanr` interface (2) in R (3). The choice of a Bayesian modeling framework allow us to naturally handle missing data in a wide class of models. This was important as it enabled us to model data from the project as *intention-to-treat* with the uncertainty associated with missing values incorporated into the posterior estimates.

A general workflow was adopted where models were incrementally built and tested for validity through simulations (4). First, implications of prior distributions were evaluated using prior-predictive simulations. Priors were selected to constrain model predictions and parameter values to *a priori* plausible ranges. This approach encourages the use of weakly informative priors on model parameters, which regularize models and shrink estimates and predictions without distorting the influence of the data on inference (5–7). In relevant cases, synthetic data were used to confirm that model inferences were not biased by model misspecification. Samples from the joint posterior distribution were used to create posterior predictive checks, in which model implications are simulated and compared with characteristics of the observed data (7). Selected models were also assessed for inferential capabilities using synthetic data simulations where parameter values are known. The sensitivity of model specifications to inference was assessed by comparing predictions across different models.

All sampling from prior and posterior distributions were done using the Hamiltonian Monte Carlo (HMC) no-U-turn sampler (Stan Reference Manual: MCMC sampling). Performance of the HMC algorithm was assessed through chain convergence statistics (R-hat) and visually inspecting converging and stationary sampling across multiple HMC chains.

Descriptive data are presented as mean and standard deviation (SD). Inferential statistics are presented as full posterior densities, and percentile intervals (PI), or probability statements derived from full posterior distributions.

2.2 Description of the data set and participants

The overall objective of the present analysis is to determine the effect of krill oil (sometimes referred to as N-3) supplementation in combination with resistance training on glucose metabolism and indicators of metabolic health. Outcome variables in the analysis are:

- Whole blood fatty acid composition, including omega-3 and omega-6 fatty acids and the omega-3 index
- Blood glucose in response to a oral glucose tolerance-test
- Metabolic syndrome (derived from sub-scores)
  - Triglycerides
  - Waist circumference
  - Blood pressure
  - HDL Cholesterol
  - Fasting glucose
- Body mass index
- Relative fat mass
- HOMA2-IR
- Adiponectin
- LDL Cholesterol
- HbA1c

At enrollment, participants were randomized into two groups, an active treatment-group (krill-oil fatty acids, 1 mg day^{-1}) and a placebo control group (Placebo; Figure 2.1, Table 2.1).

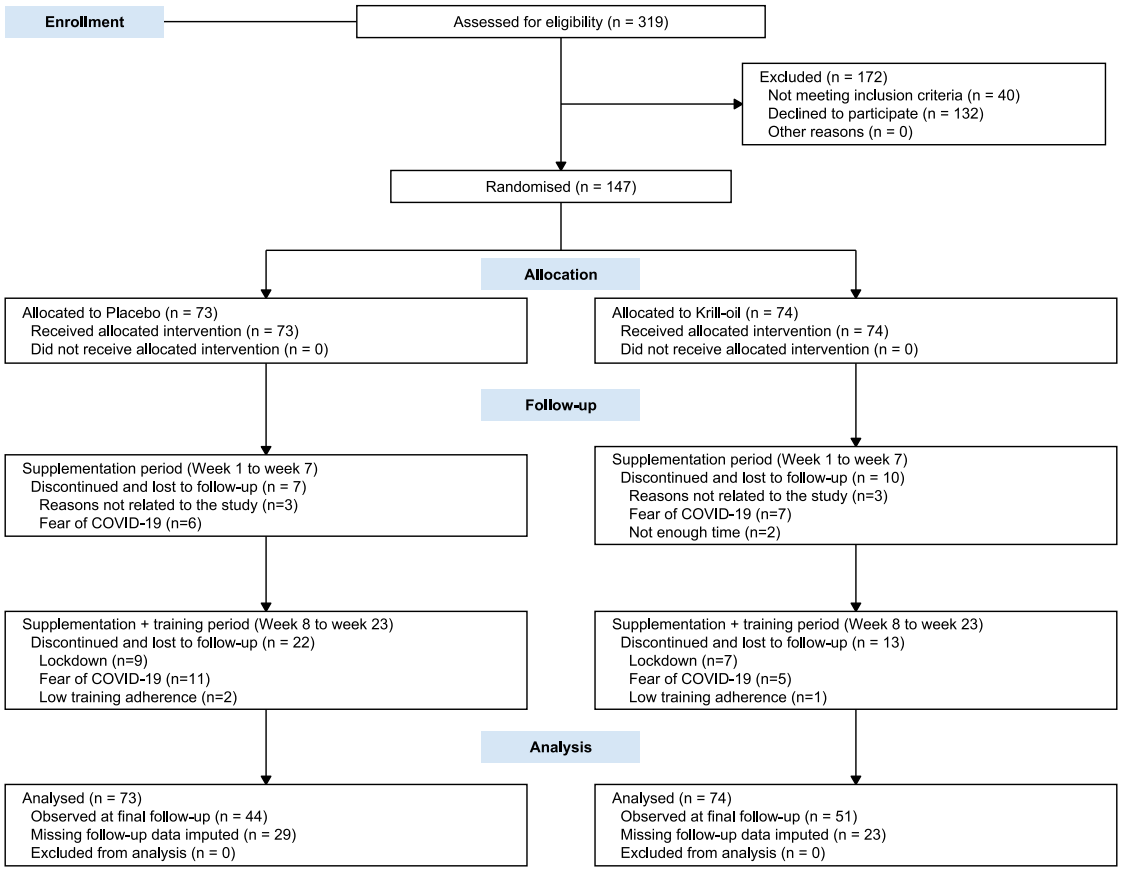


Figure 2.1: Flow of participants through the trial, presented according to the CONSORT 2025 statement. All randomized participants were included in the analysis of outcomes, with missing follow-up data imputed.

Table 2.1: Participant characteristics.

	N-3(♂ n = 32; ♀ n = 42)	Placebo(♂ n = 35; ♀ n = 38)
	Male	
Age (years)	47.3 (7.7) [29.0 - 60.0]	48.5 (5.8) [36.0 - 59.0]
Stature (cm)	180.5 (8.1) [153.3 - 193.2]	180.9 (6.4) [165.2 - 191.7]
Body mass (kg)	95.5 (17.1) [68.6 - 151.9]	97.6 (17.6) [71.2 - 142.2]
Body mass index (kg×m^{-2})	29.4 (5.1) [20.5 - 45.1]	30.0 (6.1) [23.1 - 52.1]
	Female	
Age (years)	44.5 (8.1) [30.0 - 59.0]	46.6 (9.4) [23.0 - 60.0]
Stature (cm)	168.9 (6.0) [158.0 - 181.9]	167.6 (6.4) [152.1 - 179.9]
Body mass (kg)	83.6 (16.5) [57.7 - 135.7]	79.3 (14.2) [52.9 - 113.8]
Body mass index (kg×m^{-2})	29.4 (5.8) [18.4 - 44.9]	28.2 (4.8) [20.4 - 39.9]

Data are presnted as Mean (SD) [Min-Max]

Participants were given krill oil or placebo in a supplementation-only period followed by a lead-in and combined period (supplementation and resistance training; Figure 2.2)

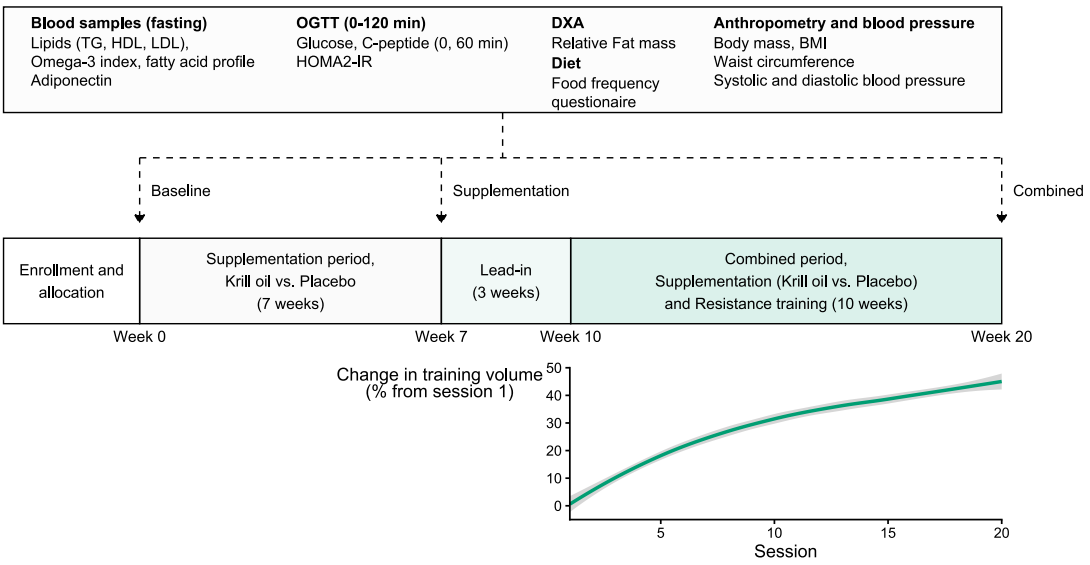


Figure 2.2: Study design. After baseline testing (week 0), participants received krill oil (N-3) or placebo for 7 weeks, followed by 13 weeks of supervised resistance training, comprising a 3-week familiarization period (weeks 8–10) and 10 weeks of progressive resistance training (weeks 11–20), with supplementation continued throughout. The measurements listed were taken at each of the three time points: baseline (week 0), after supplementation alone (end of week 7) and after the combined period (end of week 20). The lower panel shows the change in training volume across the 20 training sessions, relative to session 1 and summed across all upper- and lower-body exercises. Line and shaded area show the smoothed trend with 95% confidence interval. BMI, body mass index; DXA, dual-energy X-ray absorptiometry; HDL, high-density lipoprotein cholesterol; HOMA2-IR, updated homeostasis model assessment of insulin resistance; LDL, low-density lipoprotein cholesterol; N-3, n-3 polyunsaturated fatty acids (krill oil); OGTT, oral glucose tolerance test; TG, triglycerides.

3. Effects of N-3 supplementation of serum fatty acids

Fatty acids, including including EPA (C205n3) and DHA (226n3) where measured in red blood cells. We will assess the effect of krill oil (N-3) supplementation on group level fatty acid profiles. As preliminary models indicated heteroscedasticity of residuals, log-transformed and standardized data was used in modelling. This allows for the same weakly regularizing priors for all fatty acids (FA). In addition to each fatty acid, the Omega 3 index (8) is included in these models as the EPA and DHA content relative to the total fatty acid content of red blood cells. The omega 3 index is included in models from model `m3` (see below).

The models were designed to estimate differences between treatments (TX) at post-treatment time-points (T2 and T4) using pre-treatment ($\text{Pre}_{i[\text{FA}]}$) values as a covariate. The intercept in the model ($\alpha_{[\text{FA, TI, TX}]}$) gives us mean estimates for fatty-acid FA, time-point TI, and treatment TX.

$$
\begin{aligned}
y_{i[\text{FA}]} &\sim \text{Normal}\ (\mu_{i[\text{FA}]}, \sigma_{\text{FA}}) \\
\mu_{i[\text{FA}]} &= \alpha_{[\text{FA, TI, TX}]} + \beta_{\text{FA}} \times \text{Pre}_{i[\text{FA}]} \\
\alpha_{[\text{FA, TX}]} &\sim \text{Normal}\ (0, 1) \\
\beta_{\text{FA}} &\sim \text{Normal}\ (1, 1) \\
\sigma_{\text{FA}} &\sim \text{Exponential}\ (1)
\end{aligned}
\tag{3.1}
$$

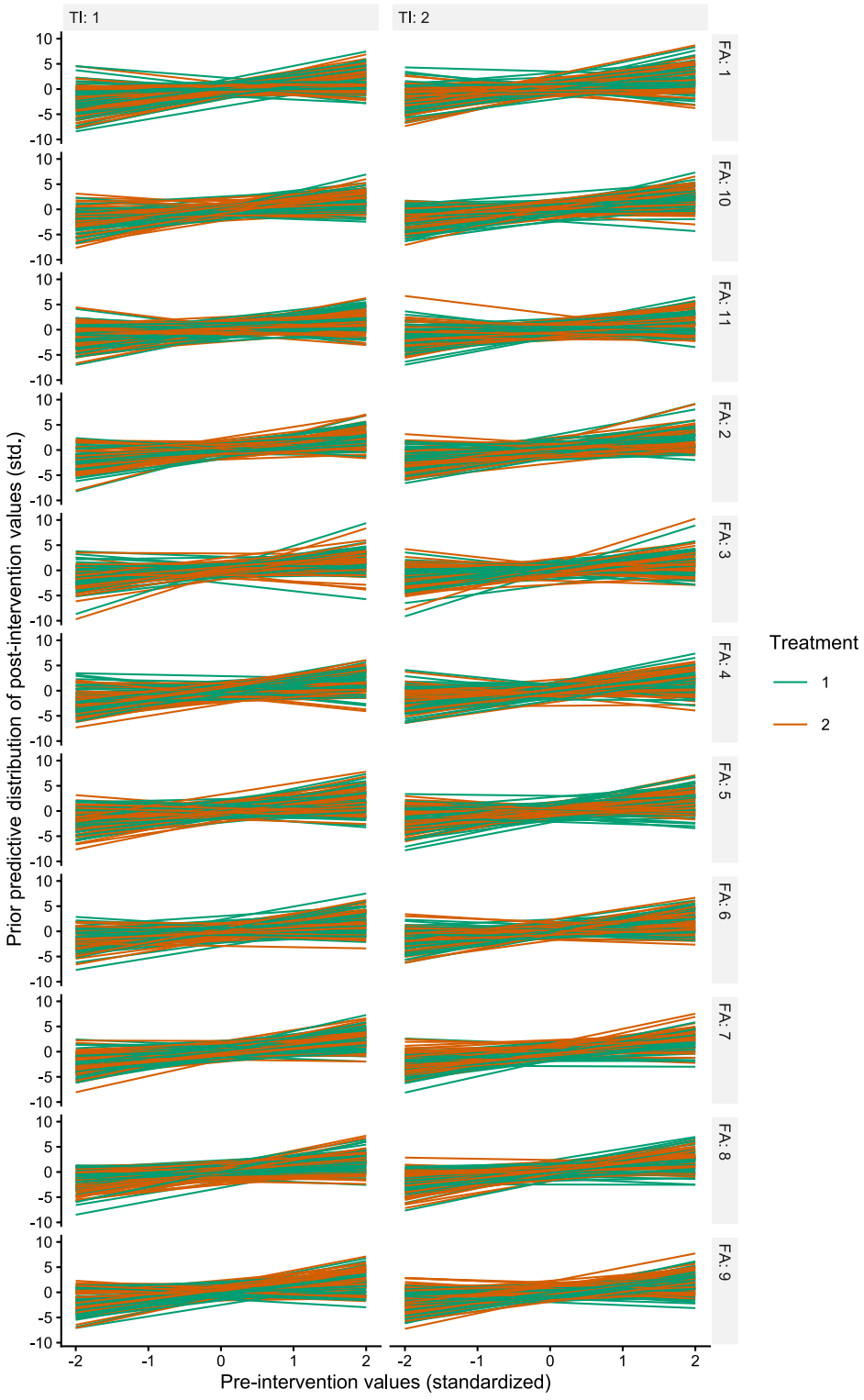


Figure 3.1: Prior predictive check for target-wise models of post-treatment fatty acid concentrations conditional on pre-treatment values. The model expects log-transformed standardized values and the priors suggest no difference between treatment groups and a proportional relationship between th pre-intervention values and post-intervention measurements.

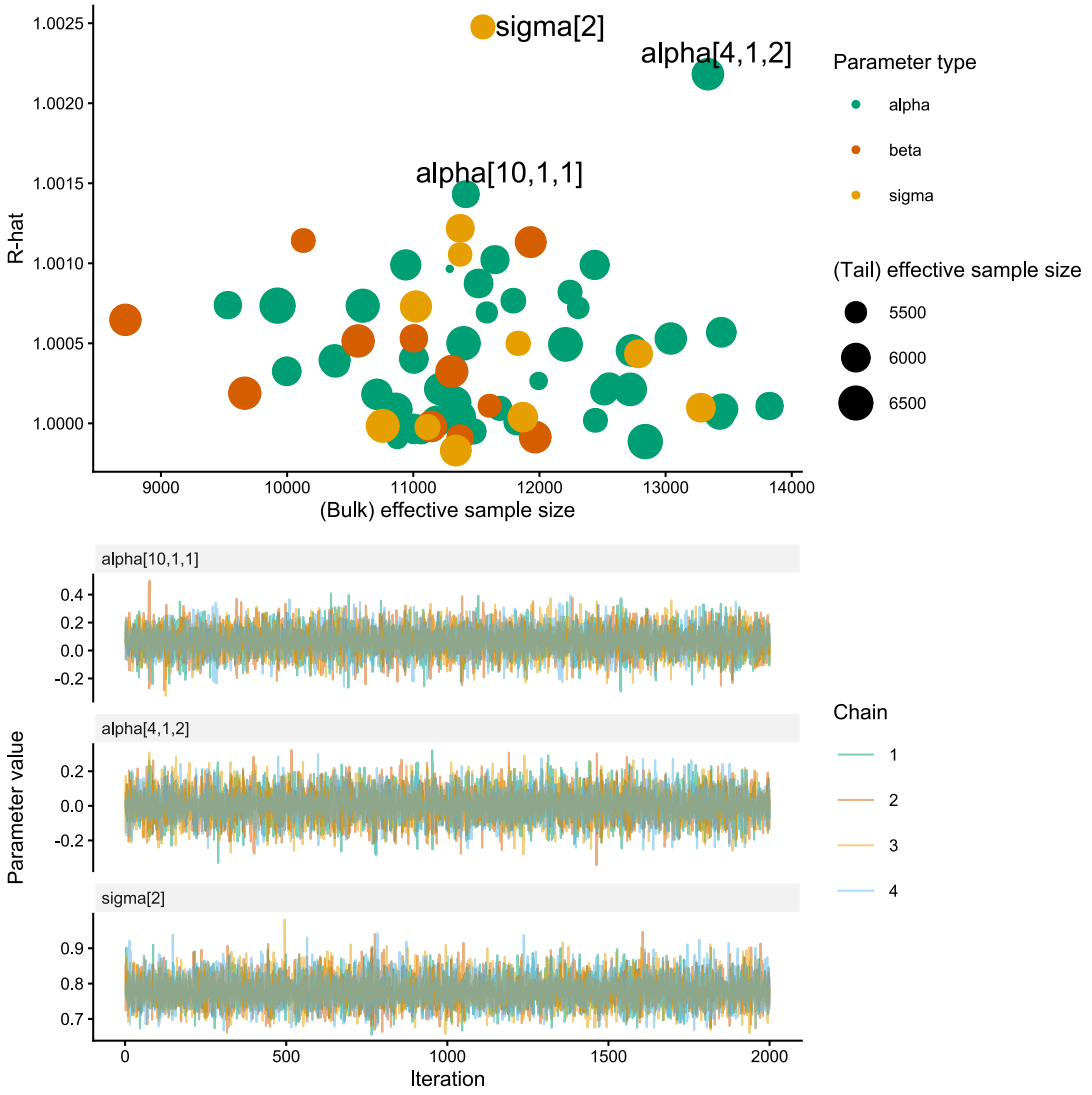


Figure 3.2: Model performance metrics from Model 2.

The first model uses only complete-cases. The prior predictive check confirms that the model expects eleven fatty acids, a positive relationship between pre-intervention values and post-treatment values, and no differences in treatments (Figure 3.1). Sampling from the posterior was non-problematic as R-hat values approached 1 and chains mixed and converged well (Figure 3.2).

Posterior predictive checks showed satisfactory patterns between replicated data (yrep; thin lines, one per draw in Figure 3.3) and observed data (thick lines in Figure 3.3).

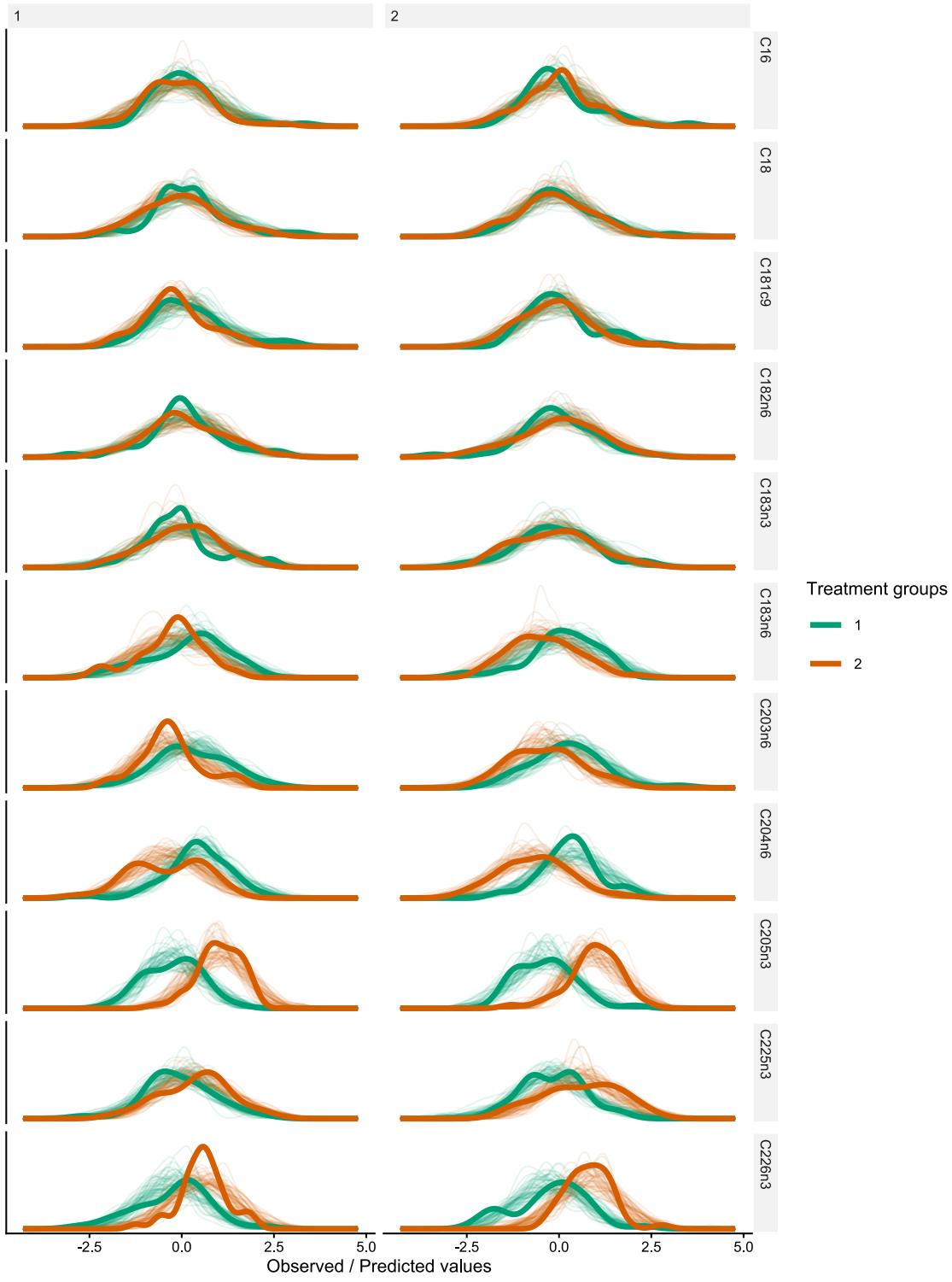


Figure 3.3: Posterior predictive check of Model 2.

Treatment with N-3 fatty acids robustly increased whole blood fatty acid composition levels of C205n3, C225n3 and C226n3 fatty acids compared to placebo (Figure 3.4 and Figure 3.5). Post-treatment values were lower in the N-3 group for C181c9, C183n6, C203n6 and C204n6 with high probability at both T2 and T4 (Figure 3.5).

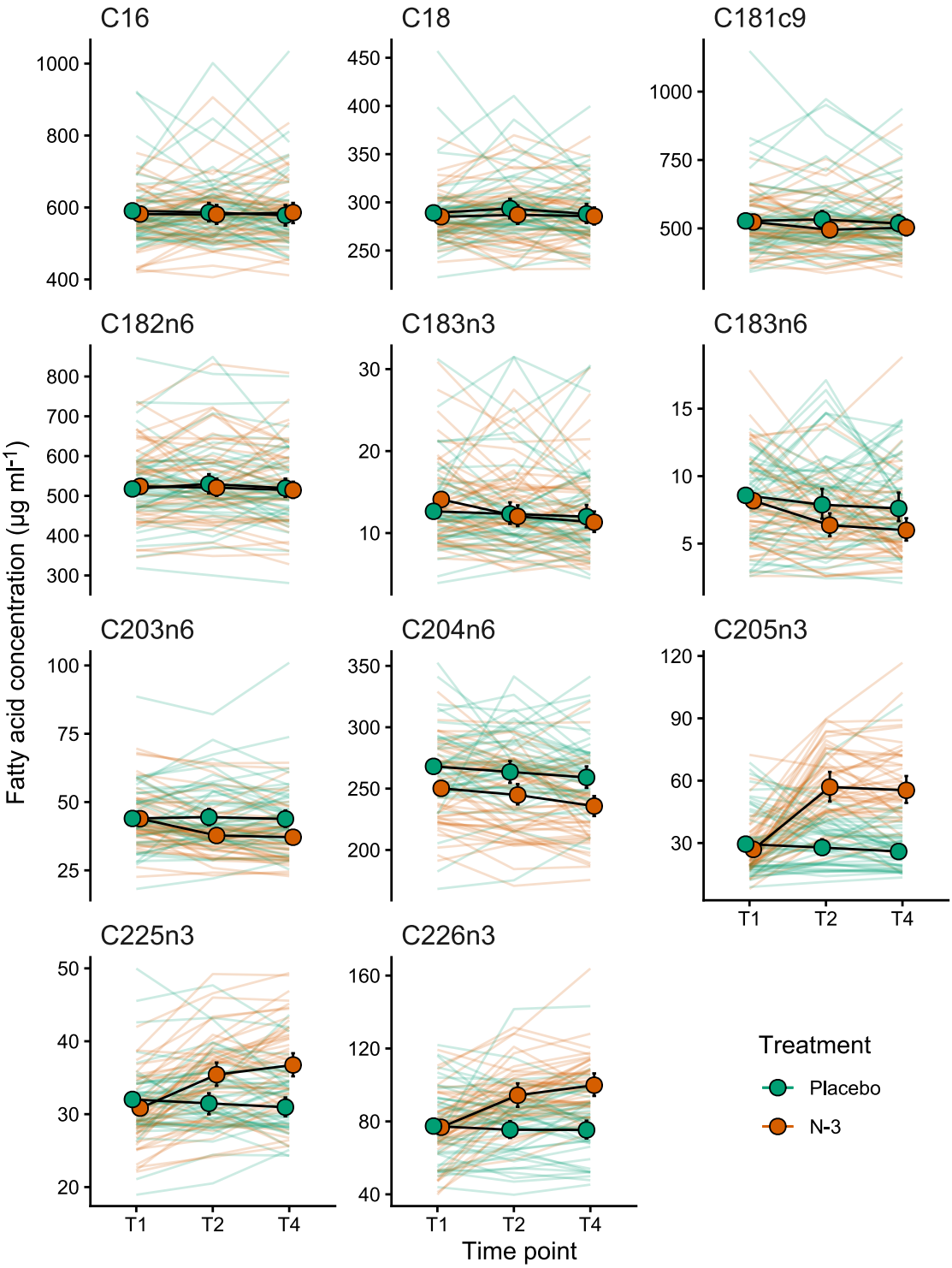


Figure 3.4: Predictions of fatty acid concentration in response to treatment. Pre-intervention values are means per treatment group, post-intervention means are predicted from the model with 99% PI.

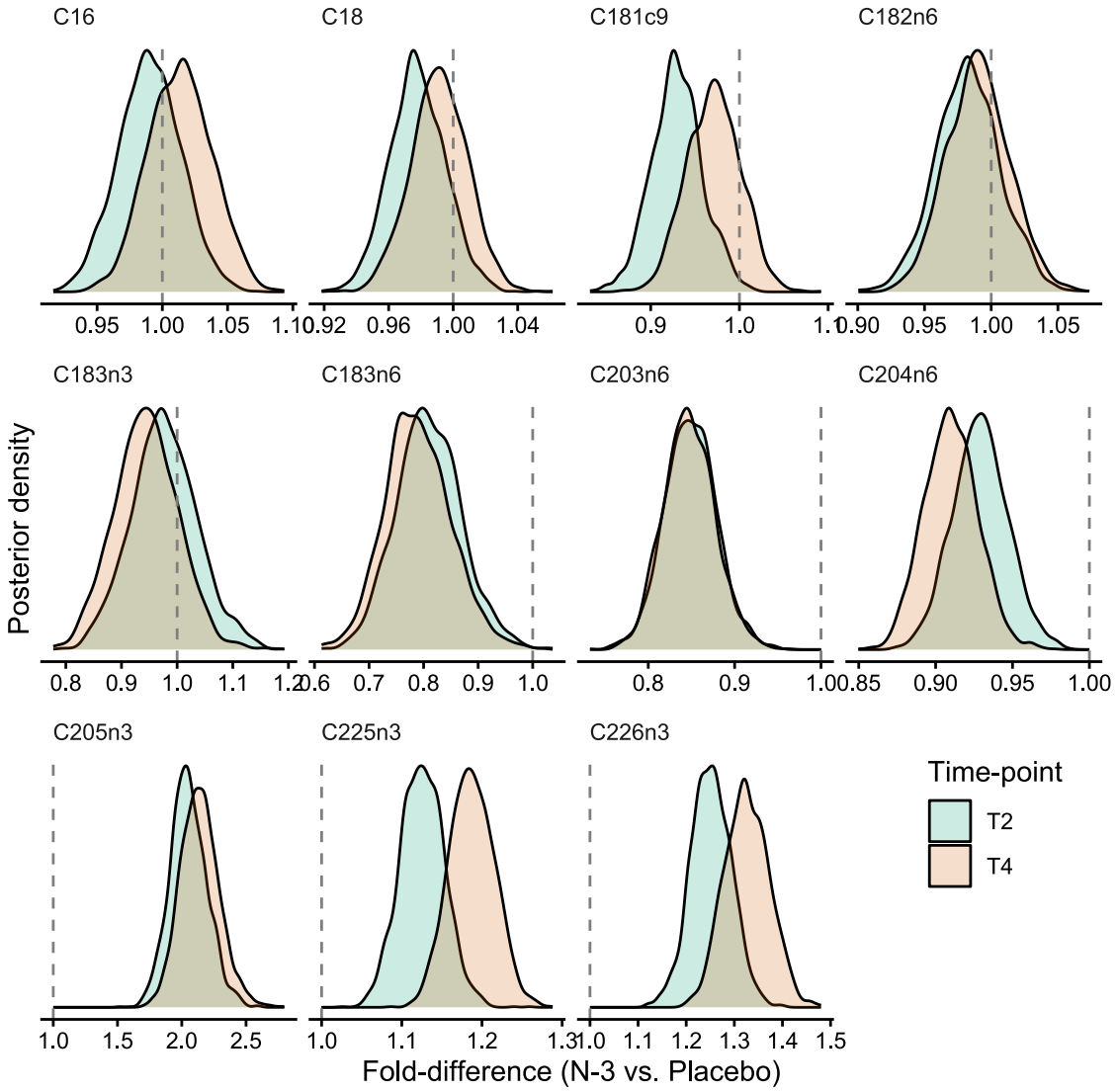


Figure 3.5: Posterior densities for differences between treatment groups in fatty acid concentrations.

Extending the first model with missing data imputations allows us to estimate the effect of treatment on fatty acid profiles with intention-to-treat (ITT) by averaging over the uncertainty associated with the missing values. Missing data imputation is done on post-treatment scores only with the total sample size being n = 143 with 14 and 36% of participants missing at time points T2 and T4, respectively.

The imputation mechanism replaces missing values with parameters estimated from observed data (pre-treatment values and treatment allocation) through the likelihood.

The imputation model is defined as:

$$
\begin{aligned}
y^{\text{all}}_{i[\text{FA}]} &\sim \text{Normal}\ (\mu_{i[\text{FA}]}, \sigma_{\text{FA}}) \\
\mu_{i[\text{FA}]} &= \alpha_{[\text{FA, TI, TX}]} + \beta_{\text{FA}} \times \text{Pre}_{i[\text{FA}]} \\
y^{\text{all}}_{i[\text{FA}]} &= \begin{cases} y^{\text{obs}}_{i[\text{FA}]} \text{ if observed} \\ y^{\text{mis}}_{i[\text{FA}]} \text{ if missing} \end{cases} \\
y^{\text{mis}}_{i[\text{FA}]} &\sim \text{Normal}\ (0, 2) \\
\alpha_{[\text{FA, TX}]} &\sim \text{Normal}\ (0, 1) \\
\beta_{\text{FA}} &\sim \text{Normal}\ (1, 1) \\
\sigma_{\text{FA}} &\sim \text{Exponential}\ (1)
\end{aligned}
\tag{3.2}
$$

Prior predictive simulations confirm the general structure of the model (Figure 3.6). Model performance metrics from posterior sampling also indicates a functioning model (Figure 3.7).

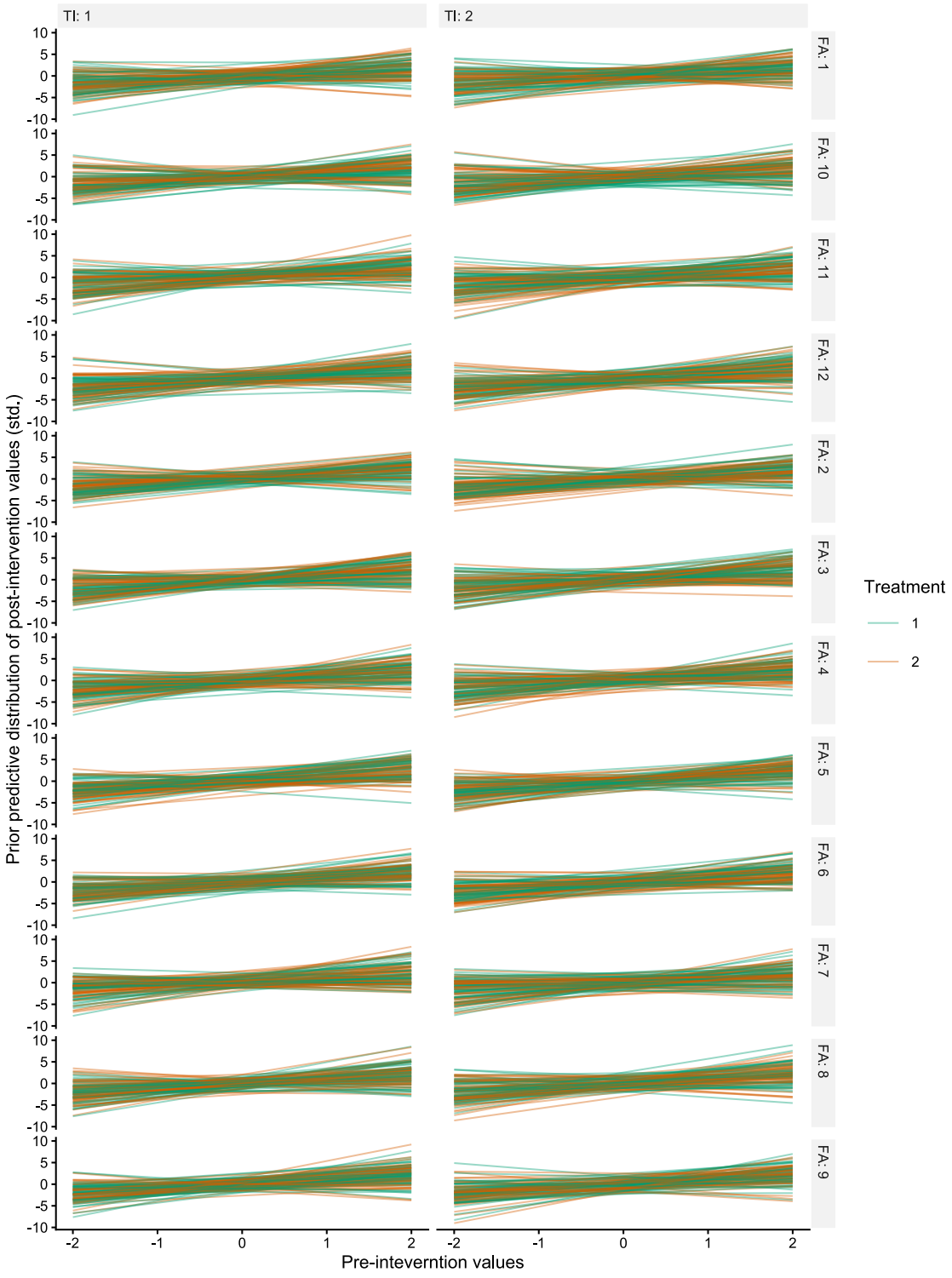


Figure 3.6: Prior predictive check model 3 (data imputation).

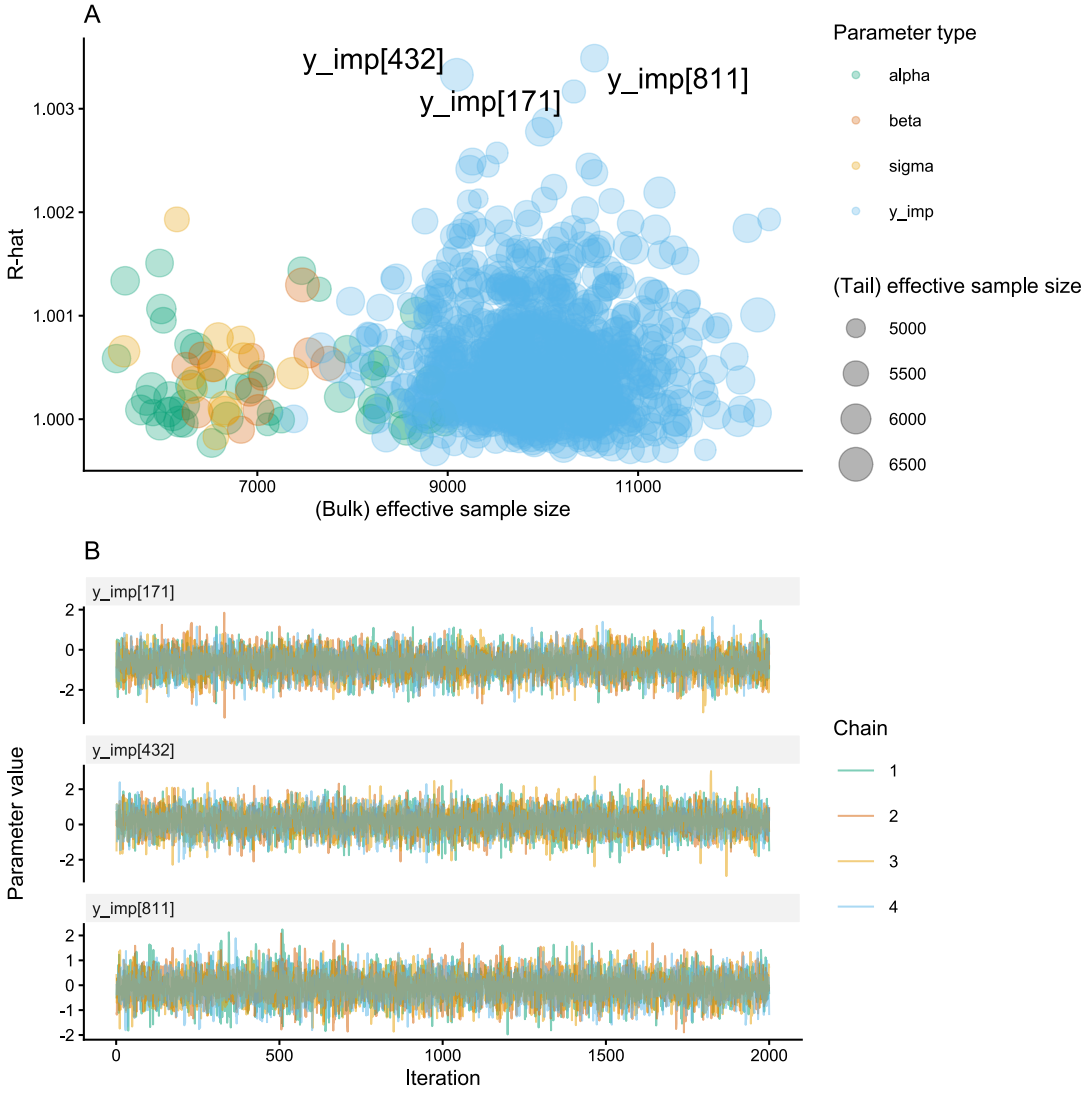


Figure 3.7: Model performance metrics model 3.

Posterior predictive checks indicated that the model capture general patterns in the data (Figure 3.8).

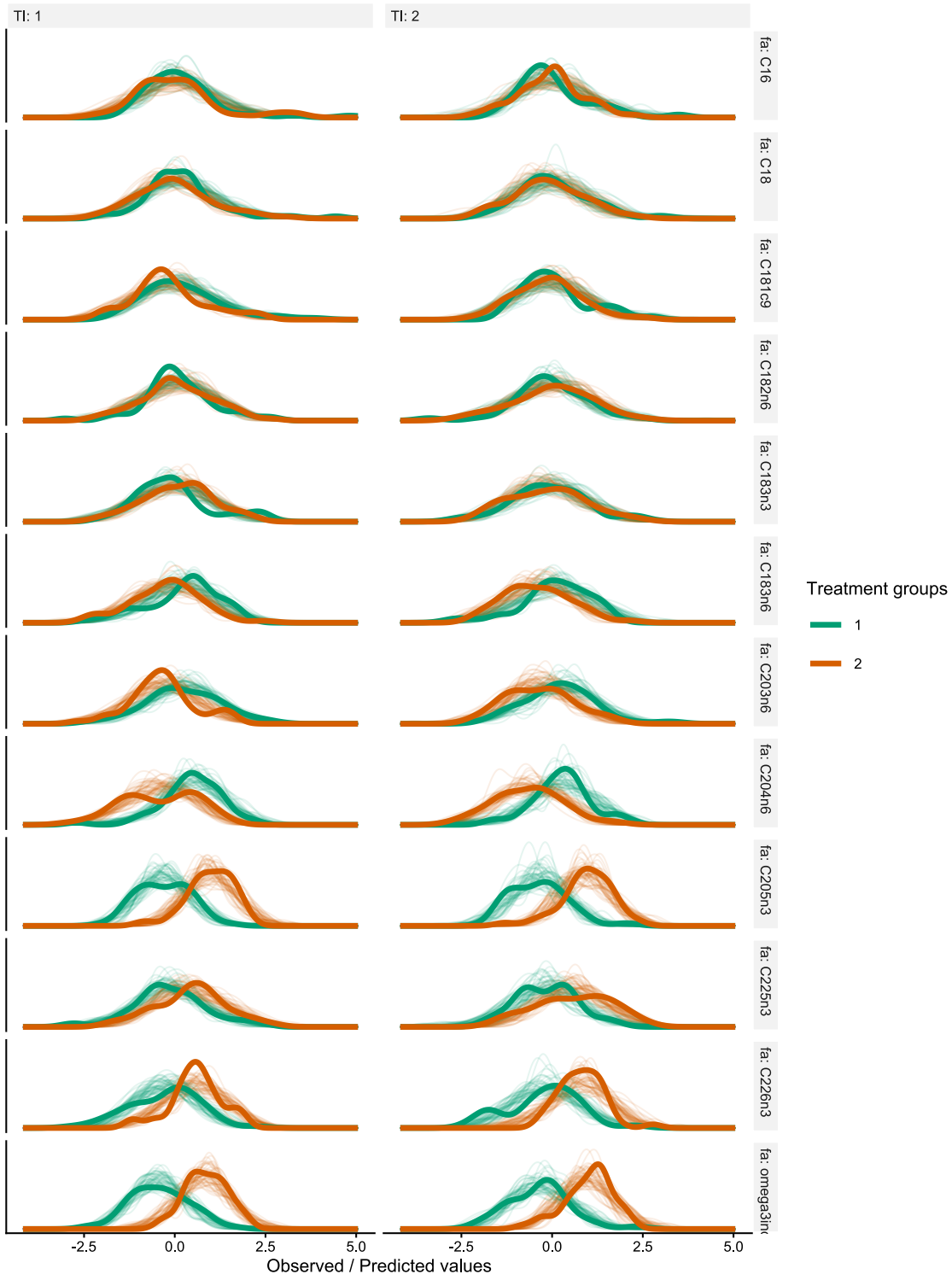


Figure 3.8: Posterior predictive check of Model 3. Thin lines represent model simulations (n = 50), thick lines represent observed data.

The imputation model generally agreed with the complete-case analysis on the effect of treatment of serum fatty acid profiles (Figure 3.9, Figure 3.10, Figure 3.11, and Figure 3.12).

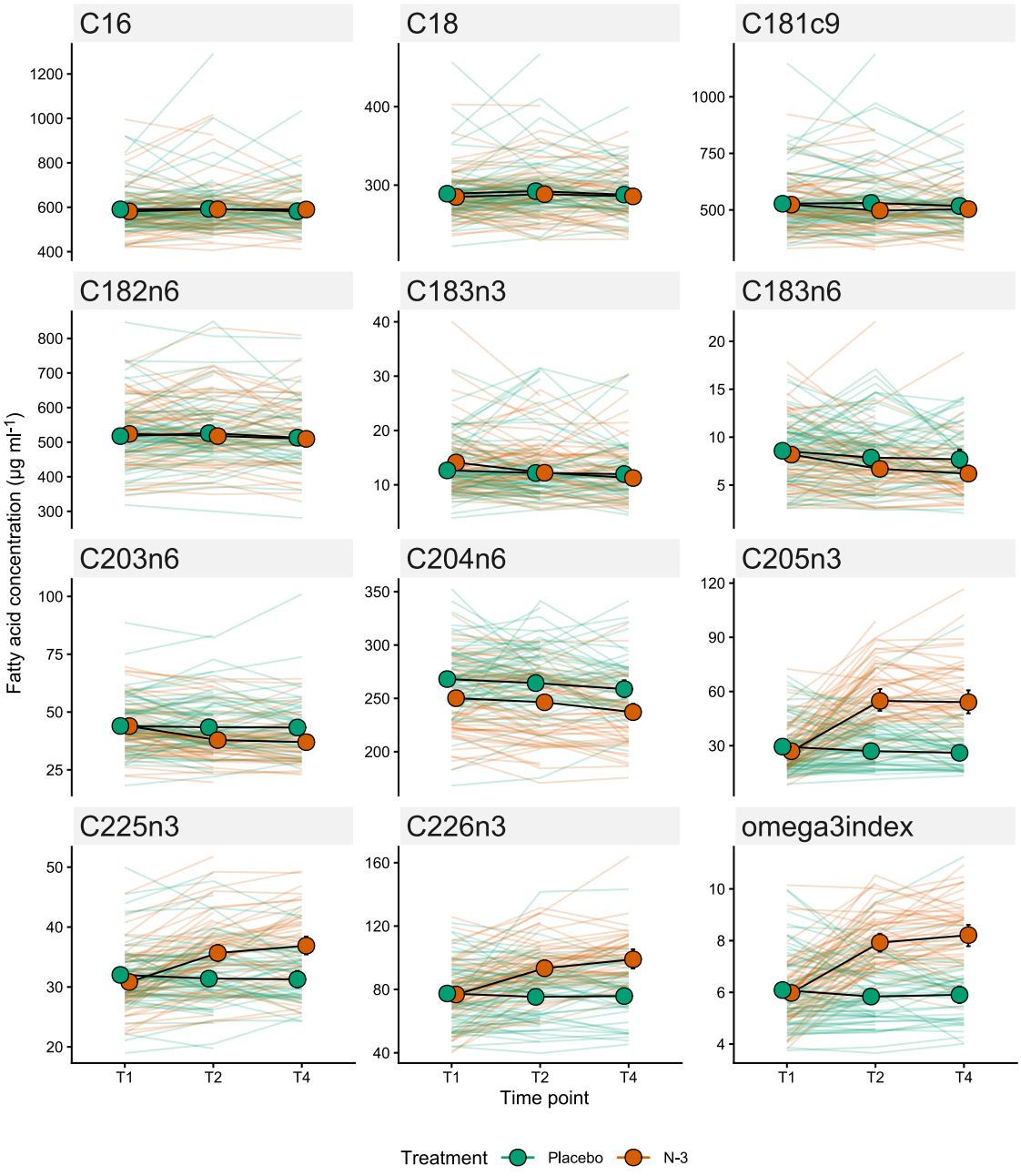


Figure 3.9: Model predictions. Pre-intervention values are observed means, post-treatment (T2, T4) values are model predictions conditional on pre-intervention value with 99% percentile intervals. The omega-3 index is EPA and DHA content of erythrocytes expressed as a percent of total identified fatty acids.

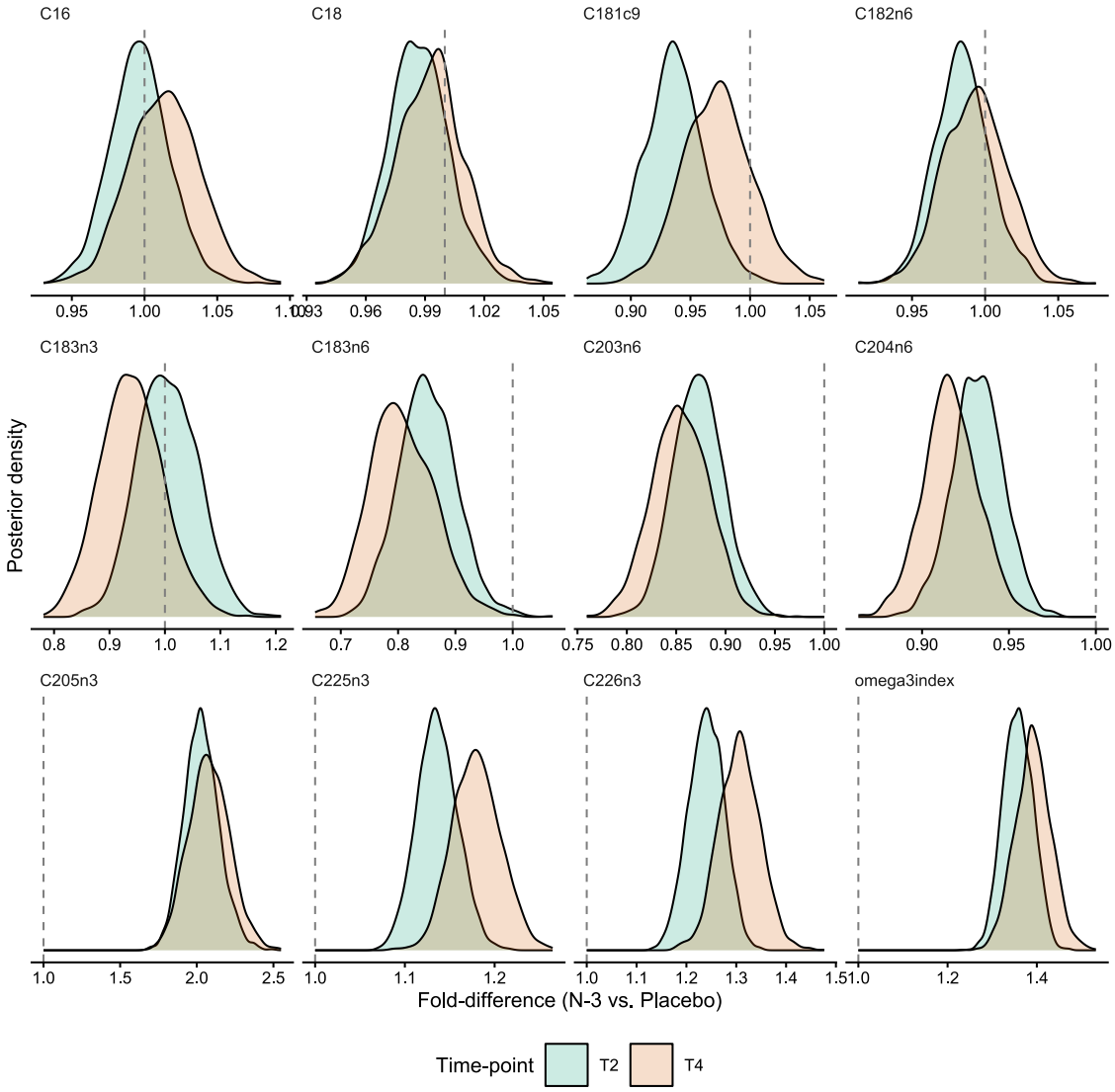


Figure 3.10: Posterior densities for differences between treatment groups in fatty acid concentrations (data imputation model).

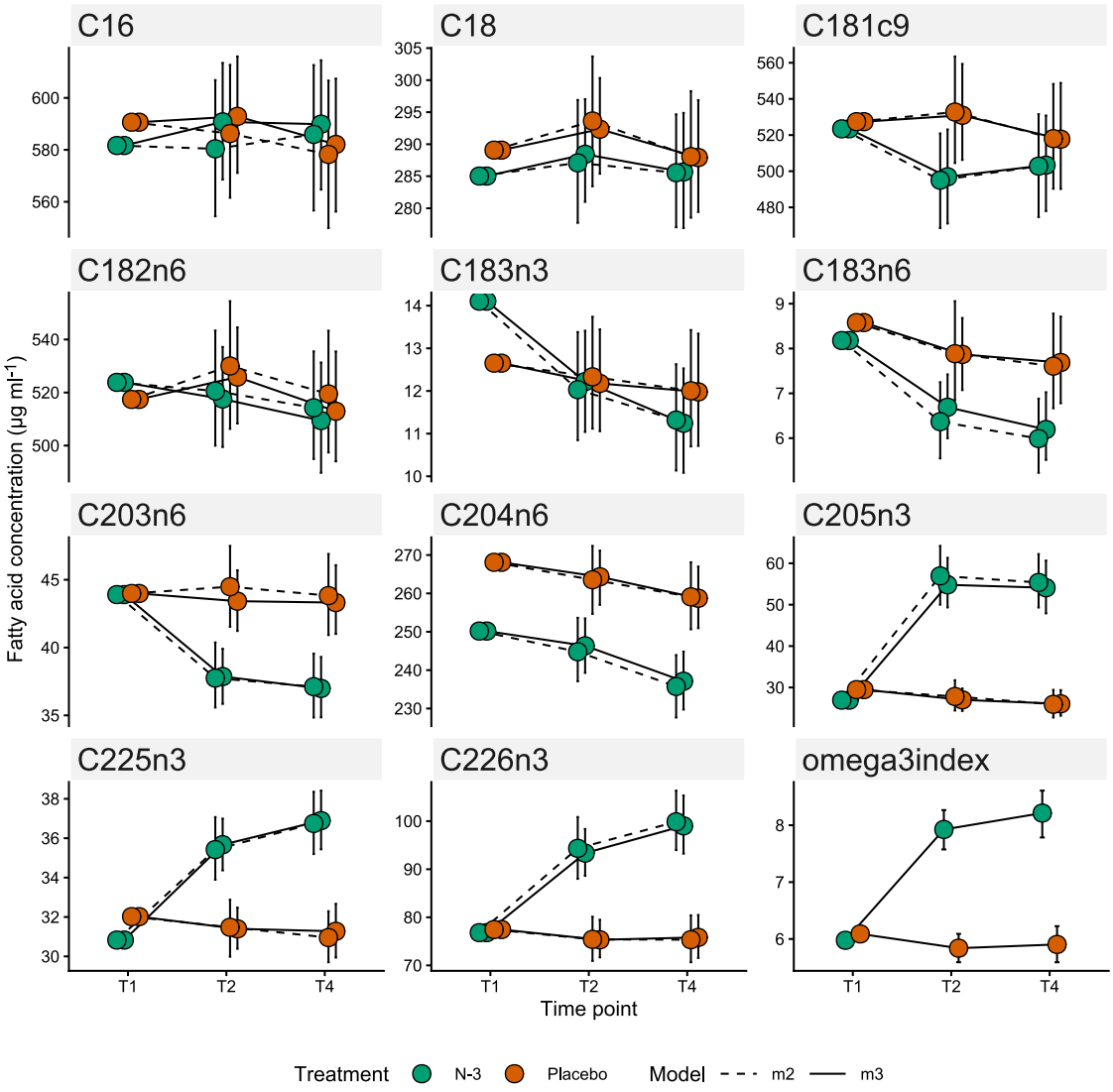


Figure 3.11: Posterior predictions for fatty acid concentrations in complete-case analysis (model m2) and data imputation model (model m3).

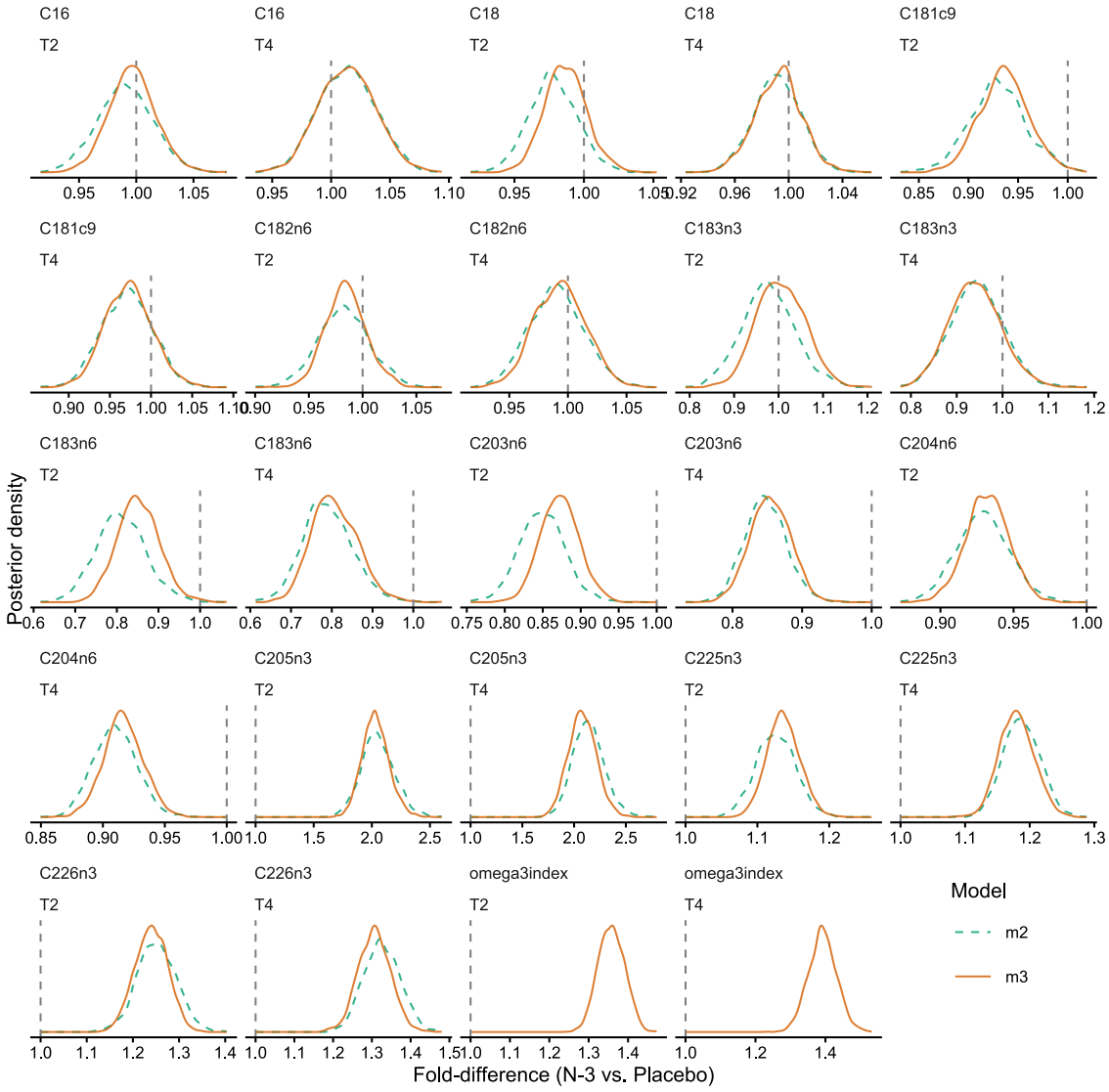


Figure 3.12: Comparing inference from complete-case and imputation models.

3.0.1 Model expansion: Heterogeneous variance

Model `m3` does not enable treatment- and time-wise heterogeneous variation. In model `m4` we therefore add $\sigma_{[\text{FA, TX}]} \sim \text{Exponential}\ (1)$, in model `m5` we add $\sigma_{[\text{FA, TI}]} \sim \text{Exponential}\ (1)$, and model `m6` adds $\sigma_{[\text{FA, TI, TX}]} \sim \text{Exponential}\ (1)$. Model performance metrics confirmed that all three extended models provided reliable chain convergence characteristics (Figure 3.13).

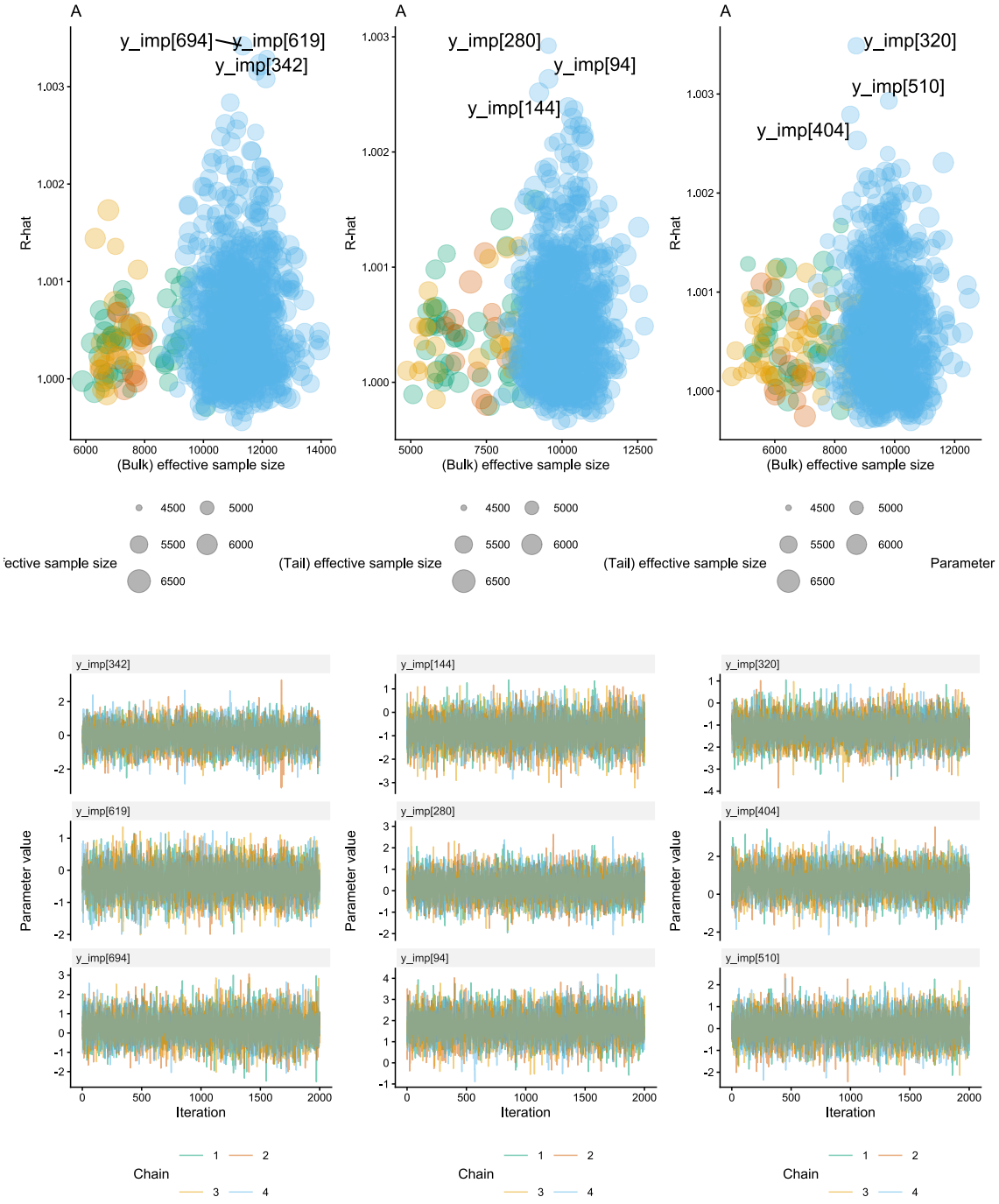


Figure 3.13: Model performance metrics models 4, 5, and 6.

Comparing the models in posterior predictive simulations of condition-specific (time, treatment and fatty-acid) SD across models indicated that no model displayed extreme deviations in replicated values compared to observed variation in the data (Table 3.1). This comparison agreed with the out of sample predictive performance where models `m3` to `m5` where comparable. Model `m6`, the most heavily parameterized model showed four unreliable Pareto k values but was also slightly worse in in out of sample predictions (Table 3.2). Together this suggest that model `m4` is the preferred model with model `m6` being the worst model.

Table 3.1: Posterior predictive p-values comparing models m3 to m6.

Model	Proportion extreme p(< 0.05 | > 0.95)	Median p	Min p	Max p
m3	0.104	0.352	0.027	0.996
m4	0	0.376	0.065	0.941
m5	0.062	0.366	0.015	0.998
m6	0.021	0.427	0.061	0.959

Table 3.2: Expected pointwise predictive density across models.

Model	ΔELPD	SE(ΔELPD)	ELPD LOO
m3	0	0	−2,575.667
m4	−0.583	6.537	−2,576.25
m5	−3.901	5.931	−2,579.568
m6	−18.138	8.559	−2,593.805

3.1 Inference

We draw inference from model `m4` regarding the effect of treatment on whole blood fatty acid profiles.

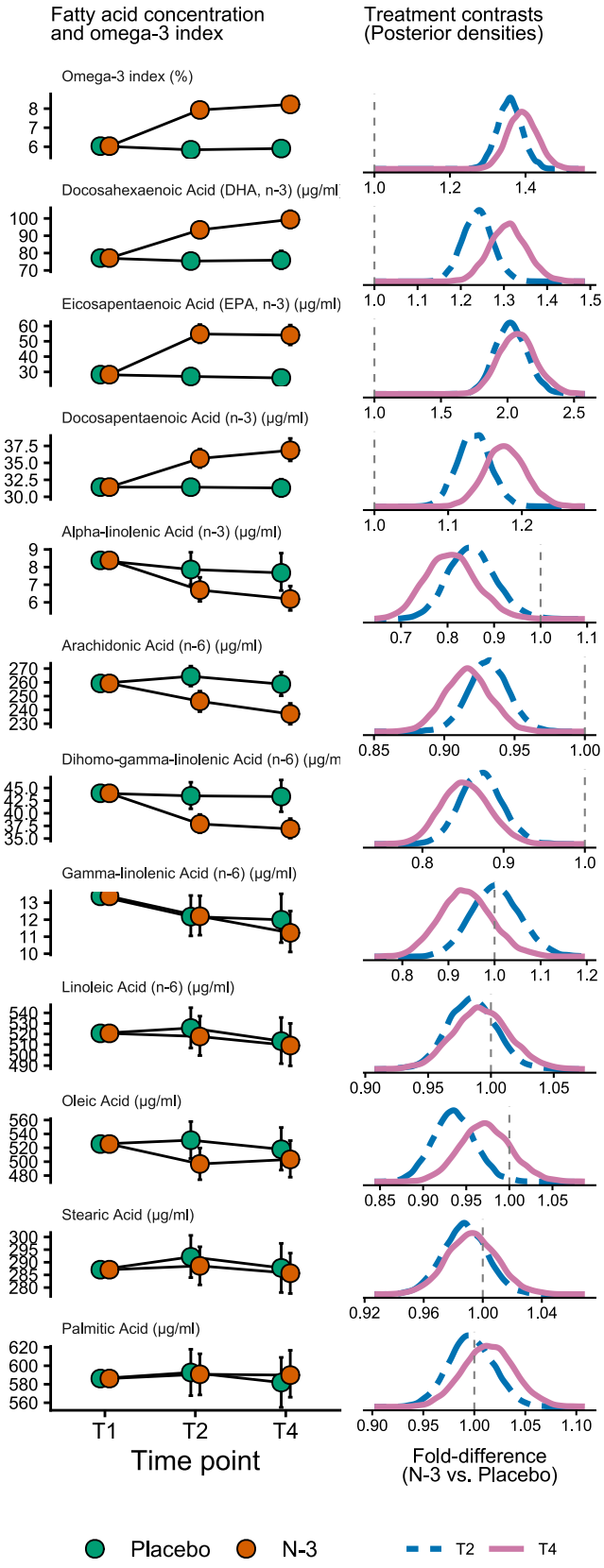


Figure 3.14: Adjusted group means and treatment contrasts of fatty acids concentrations and the omega-3 index in whole blood. Error bars in treatment-group means represents 95% CrI, posterior contrasts are displayed as the full posterior distribution ratio between N-3 to Placebo estimates at time-point T2 and T4 (1 indicating equivalent concentrations).

Fish-oil supplementation led to robust increases in Docosahexaenoic (DHA; time-point T2 N-3 vs. Placebo, 23.8% [17.2, 30.5]; T4, 31.1% [25.3, 37.7]), Eicosapentaenoic (EPA; T2, 98.8% [81.1, 127.9]; T4, 108.8% [87.4, 129.2]), and Docosapentaenoic acid (T2, 13.6% [9.6, 18.2]; T4, 17.9% [13.1, 22.8]; Figure 3.14, Table 3.3). At the same time, omega-6 fatty acids were reduced in the N-3 group compared to Placebo at both time-points (Alpha-linolenic acid T2, −14.5% [−21.8, −2.4]; Alpha-linolenic acid T4, −18.1% [−27.9, −8.2]; Arachidonic acid T2, −7.0% [−9.5, −4.6]; Arachidonic acid T4, −8.5% [−11.6, −4.9]; Dihomo-gamma-linolenic acid T2, −12.8% [−17.9, −7.3]; Dihomo-gamma-linolenic acid T4, −14.0% [−19.7, −7.6]; Figure 3.14, Table 3.3). While gamma-linolenic, linoleic, steraic and palmitic acids remained stable unchanged in N-3 compared to Placebo (Figure 3.14, Table 3.3), Oleic acid was reduced in N-3 compared to control at T2 (-6.2% [−10.3, −2.5]) but not at T4 (-2.4% [−9.1, 3.0]).

Table 3.3: Descriptive data, adjusted mean estimates, prediction intervals and treatment contrasts for fatty acids measured in whole blood before the intervention (time-point T1), after supplementation-only (T2), and after supplementation and resistance training (T4). All values are reported on the measurement scale (fatty acids, µg ml-1; omega-3 index, %).

	Time-point	Observed $\bar{x}$ (SD)		Estimate μ [95%]		Contrast [95%]
		Placebo	n-3	Placebo	n-3	n-3 vs. placebo
Omega-3 index (%)	T1	6.1 (1.5)	6.0 (1.2)			
	T2	5.9 (1.2)	8.0 (1.1)	5.8 [5.6, 6.1]	7.9 [7.7, 8.2]	2.2 [1.8, 2.5]
	T4	6.1 (1.3)	8.3 (1.3)	5.9 [5.7, 6.2]	8.2 [7.9, 8.5]	2.4 [2.0, 2.8]
Docosahexaenoic Acid (DHA, n-3)	T1	77 (20)	77 (19)			
	T2	78 (19)	95 (17)	75 [72, 79]	93 [90, 97]	18.8 [13.9, 23.5]
	T4	79 (20)	100 (16)	76 [72, 80]	99 [96, 103]	24.4 [18.7, 30.1]
Eicosapentaenoic Acid (EPA, n-3)	T1	30 (15)	27 (13)			
	T2	30 (12)	58 (18)	27 [25, 29]	55 [50, 59]	30.8 [25.3, 36.5]
	T4	29 (15)	58 (20)	26 [24, 29]	54 [49, 59]	31.0 [24.9, 37.3]
Docosapentaenoic Acid (n-3)	T1	32 (5)	31 (5)			
	T2	32 (6)	36 (6)	31 [31, 32]	36 [35, 37]	4.4 [3.0, 5.8]
	T4	32 (5)	37 (6)	31 [30, 32]	37 [36, 38]	5.7 [4.1, 7.4]
Alpha-linolenic Acid (n-3)	T1	8.5 (3.1)	8.0 (3.6)			
	T2	8.8 (3.6)	7.1 (3.3)	7.9 [7.2, 8.6]	6.7 [6.2, 7.2]	−1.1 [−2.0, −0.3]
	T4	8.2 (3.2)	6.6 (3.1)	7.7 [6.9, 8.5]	6.2 [5.7, 6.7]	−1.4 [−2.4, −0.5]
Arachidonic Acid (n-6)	T1	268 (33)	250 (34)			
	T2	273 (29)	242 (34)	264 [259, 270]	246 [241, 252]	−18.0 [−25.7, −10.3]
	T4	265 (29)	234 (31)	259 [253, 265]	237 [231, 243]	−21.5 [−29.8, −13.1]
Dihomo-gamma-linolenic Acid (n-6)	T1	44 (12)	43 (11)			
	T2	46 (13)	39 (10)	43 [41, 46]	38 [37, 39]	−5.4 [−7.9, −3.1]
	T4	45 (13)	38 (10)	43 [41, 46]	37 [35, 38]	−6.3 [−9.0, −3.6]
Gamma-linolenic Acid (n-6)	T1	13 (5)	14 (6)			
	T2	13 (7)	14 (5)	12 [11, 13]	12 [11, 13]	0.0 [−1.3, 1.3]
	T4	13 (6)	13 (6)	12 [11, 13]	11 [10, 12]	−0.8 [−2.1, 0.6]
Linoleic Acid (n-6)	T1	523 (95)	531 (91)			
	T2	536 (96)	534 (98)	526 [511, 540]	518 [504, 532]	−8.2 [−28.1, 11.7]
	T4	522 (95)	529 (103)	513 [496, 531]	509 [494, 525]	−4.2 [−27.1, 18.4]
Oleic Acid	T1	535 (139)	521 (116)			
	T2	553 (151)	507 (121)	531 [511, 552]	497 [480, 515]	−34.3 [−60.3, −8.0]
	T4	533 (121)	510 (109)	518 [495, 541]	503 [484, 523]	−14.7 [−45.1, 16.2]
Stearic Acid	T1	290 (36)	285 (29)			
	T2	296 (38)	289 (35)	292 [286, 298]	289 [283, 294]	−3.6 [−12.3, 4.9]
	T4	291 (32)	286 (31)	288 [280, 295]	286 [279, 292]	−2.1 [−11.9, 7.5]
Palmitic Acid	T1	593 (95)	577 (105)			
	T2	609 (125)	595 (124)	593 [574, 612]	591 [573, 608]	−2.2 [−28.0, 23.4]
	T4	595 (108)	584 (87)	582 [562, 603]	590 [571, 609]	8.1 [−21.0, 36.2]

4. Effects of krill oil supplementation on indications of metabolic syndrome - Introduction

4.1 Effects of N-3 supplementation on indicies of metabolic syndrome

Here we aim to model the components of the metabolic syndrome definition (9), along with related outcomes (Table 4.2). The goal is to build a model that imputes missing data to enable an intention-to-treat (ITT) analysis of all randomized participants. Imputation will also allow for assessing the probability of meeting metabolic syndrome definition thresholds in each group while accounting for uncertainty associated with missing observations. The metabolic syndrome definition requires a minimum waist circumference (central obesity) and two more threshold values among the set of criteria markers (9).

Figure 4.1 shows distributions of the variables selected for joint modeling, including variables used to define metabolic syndrome (waist circumference, raised triglycerisdes, reduced HDL-cholesterol, raised blood pressure, raised fasting glucose).

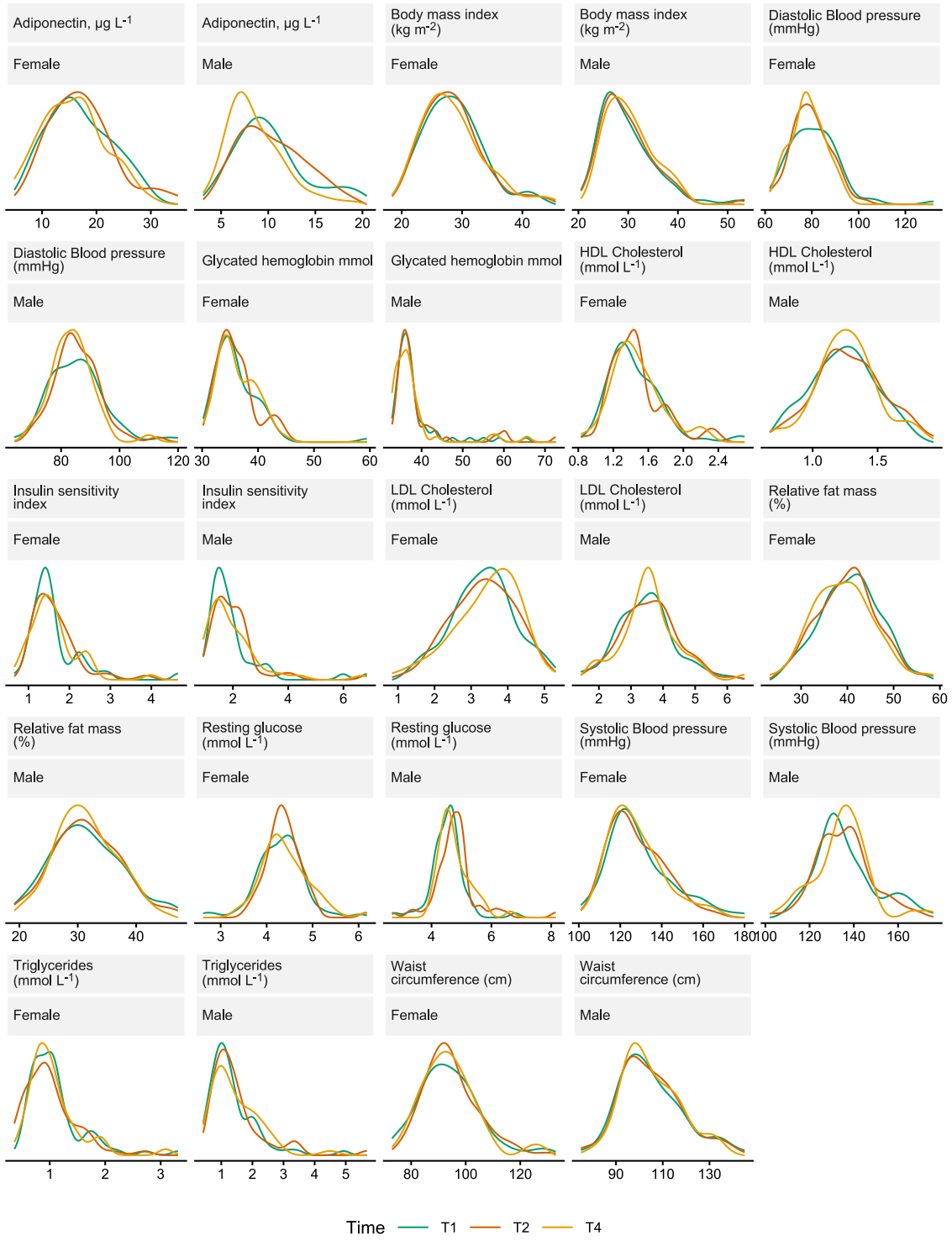


Figure 4.1: Observed data distributions of main outcomes.

Descriptive analysis of complete cases indicates variation in the proportion of participants meeting the criteria thresholds across treatment groups and post-intervention time points. Table 4.1 and Figure 4.2 show the proportions of participants who have three or more diagnostic criteria for metabolic syndrome among complete cases.

Table 4.1: Participants in the complete-case analysis of metabolic syndrome indications.

	Time		
	T1	T2	T4
	Male		
Placebo	21 (19%)	21 (29%)	21 (38%)
Krill oil	19 (32%)	19 (16%)	19 (32%)
	Female		
Placebo	16 (12%)	16 (12%)	16 (19%)
Krill oil	21 (19%)	21 (10%)	21 (14%)
	Total		
Placebo	37 (16%)	37 (22%)	37 (30%)
Krill oil	40 (25%)	40 (12%)	40 (22%)

Data are presented as n (Metabolic syndrome proportion).Time T1, Baseline;T2 Supplementation; T4, Combined.

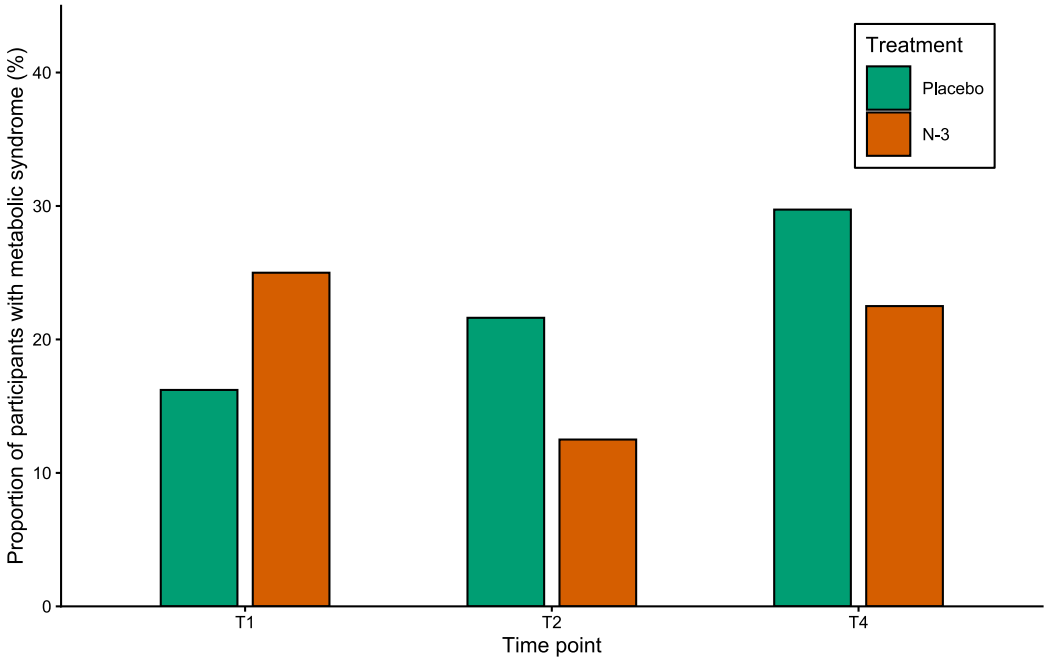


Figure 4.2: Proportion of participants with metabolic syndrome (at least three criteria thresholds).

Table 4.2: Components of the metabolic syndrome diagnostic criteria and related measurements used in modelling.

Subscore	Threshold
Waist circumference*	> 102 cm for men, > 88 cm for women
HDL Cholesterol	< 40 mg/dL for men, < 50 mg/dL for women
Fasting glucose	100 mg/dL or higher (or taking medication)
Triglycerides	150 mg/dL or higher (or taking medication)
Blood pressure	130/85 mm Hg or higher (or taking medication)
Relative fat mass	NA
Body mass index	NA
Insulin resistance index	NA
Adiponectin	NA
LDL Cholesterol	NA
HbA1c	NA

*, required for metabolic syndrome diagnosis

Variables in the observed data are correlated within time points within variables and between variables across time points. To describe approximate variation and co-variation patterns, "naive" linear models were fitted with sex as a single predictor at time-point T1, sex, baseline values, and treatment as predictors for values at time-points T2 and T4. Residuals were extracted from the models, and estimates were plotted in Figure 4.3.

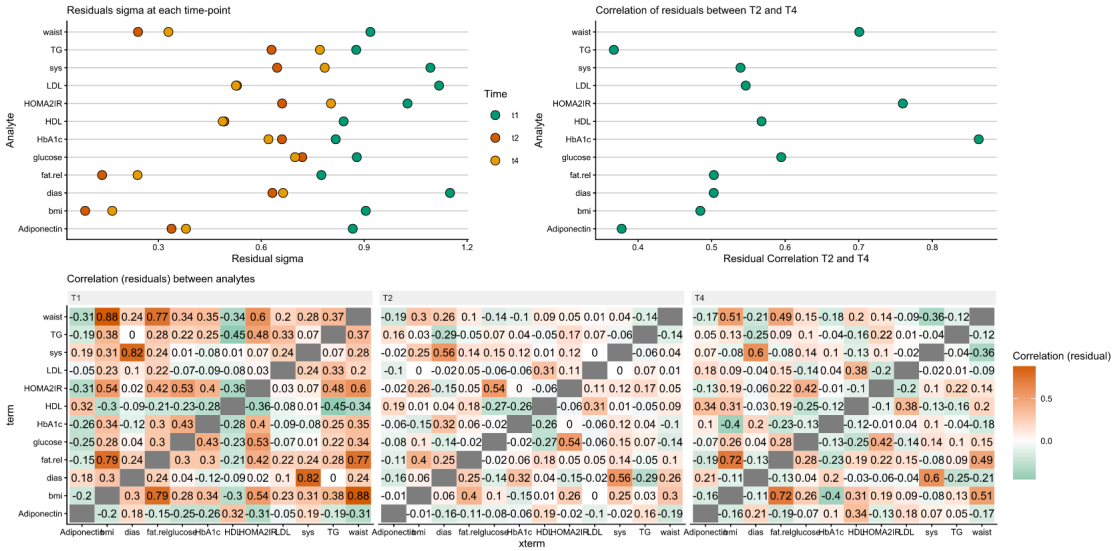


Figure 4.3: Residual variation and correlation pattern within and between variables in the observed data.

Correlation patterns among variables will be used to inform simulations (see Section 4.1.1). Correlations among variables also suggest that uncertainty associated with missing data can be reduced by exploiting the covariance among variables. The missing data patterns in the data set indicate that some missing data occurs in a single variable at a single time point whilst some participants have missing data in all variables at specific time points (Figure 4.4).

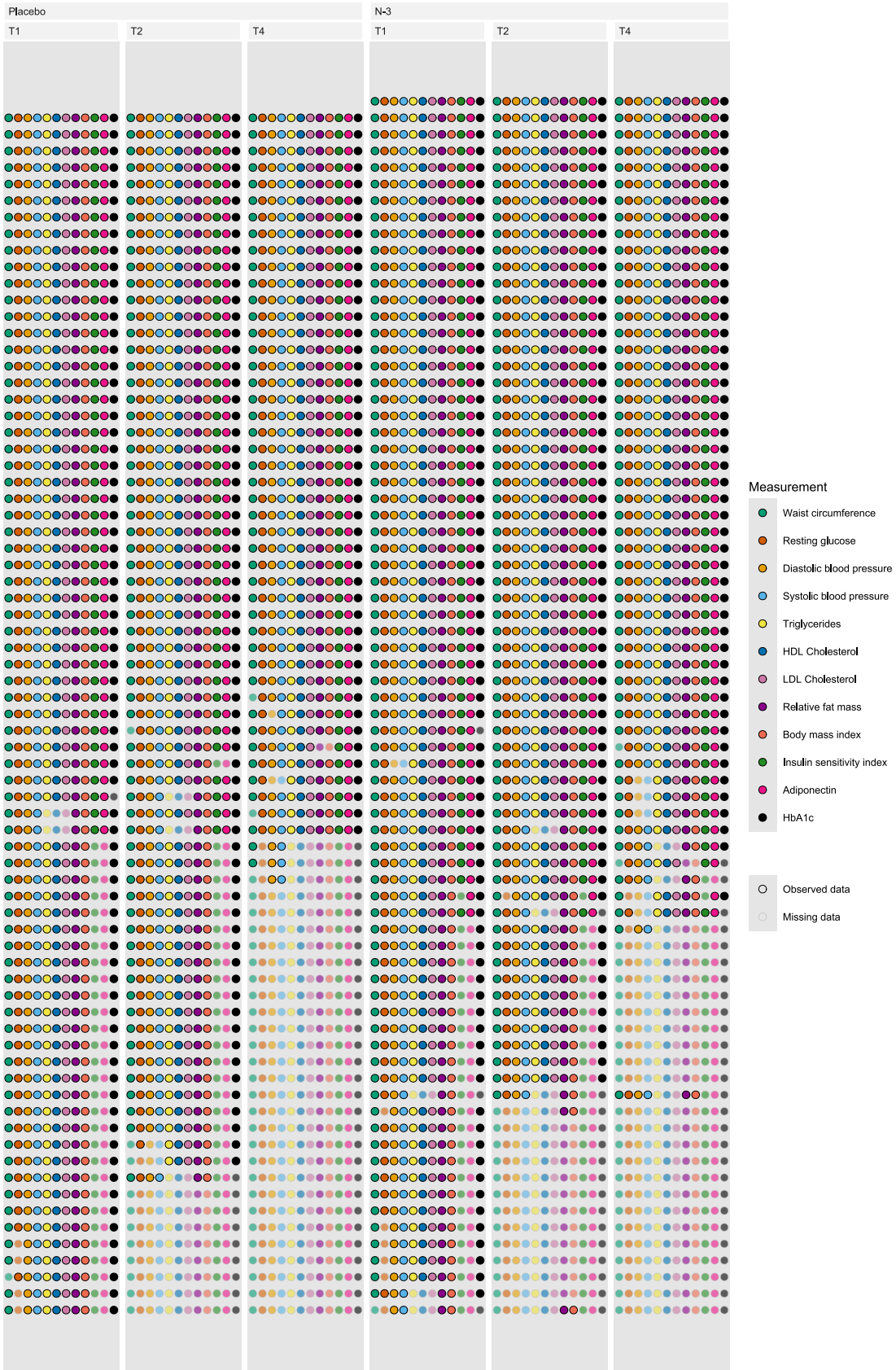


Figure 4.4: Observed and missing data across treatment groups and time-points. Rows are participants, columns are data points.

To assess if the outcome variables predict drop-out we fit a logistic regression model with drop-out (D_i) at time Time T2 or T4, coded as 0/1 as the outcome variable. The association between the standardized measurement type value ($X_{i[A]}$) and drop-out was determined while controlling for sex (S) within each treatment group (T).

$$
\begin{aligned}
D_{i[\text{Time}]} &\sim \text{Bernoulli } (p_i) \\
\text{Logit } (p_i) &= \alpha_{[A]} + \beta_{X[A]} X_{i[A]} + \beta_{S[A]} S_i + \beta_{T[A]} T_i + \beta_{TX[A]} T_i X_i \\
\alpha_{[A]} &\sim \text{Normal } (0, 0.5) \\
\beta_{X[A]} &\sim \text{Normal } (0, 0.5) \\
\beta_{S[A]} &\sim \text{Normal } (0, 0.5) \\
\beta_{T[A]} &\sim \text{Normal } (0, 0.5) \\
\beta_{TX[A]} &\sim \text{Normal } (0, 0.5)
\end{aligned}
\tag{4.1}
$$

In addition to measurements included in the metabolic syndrome diagnostic criteria (Fasting glucose, waist circumference, HDL, triglycerides, blood pressure), we also included relative fat mass, body mass index, age and insulin resistance index (HOMA2IR), LDL, and adiponectin. Prior predictive checks (Figure 4.5) confirmed that the model expects eleven measurement types in two treatment groups and that the combined priors weakly regularize away from all/non drop-out.

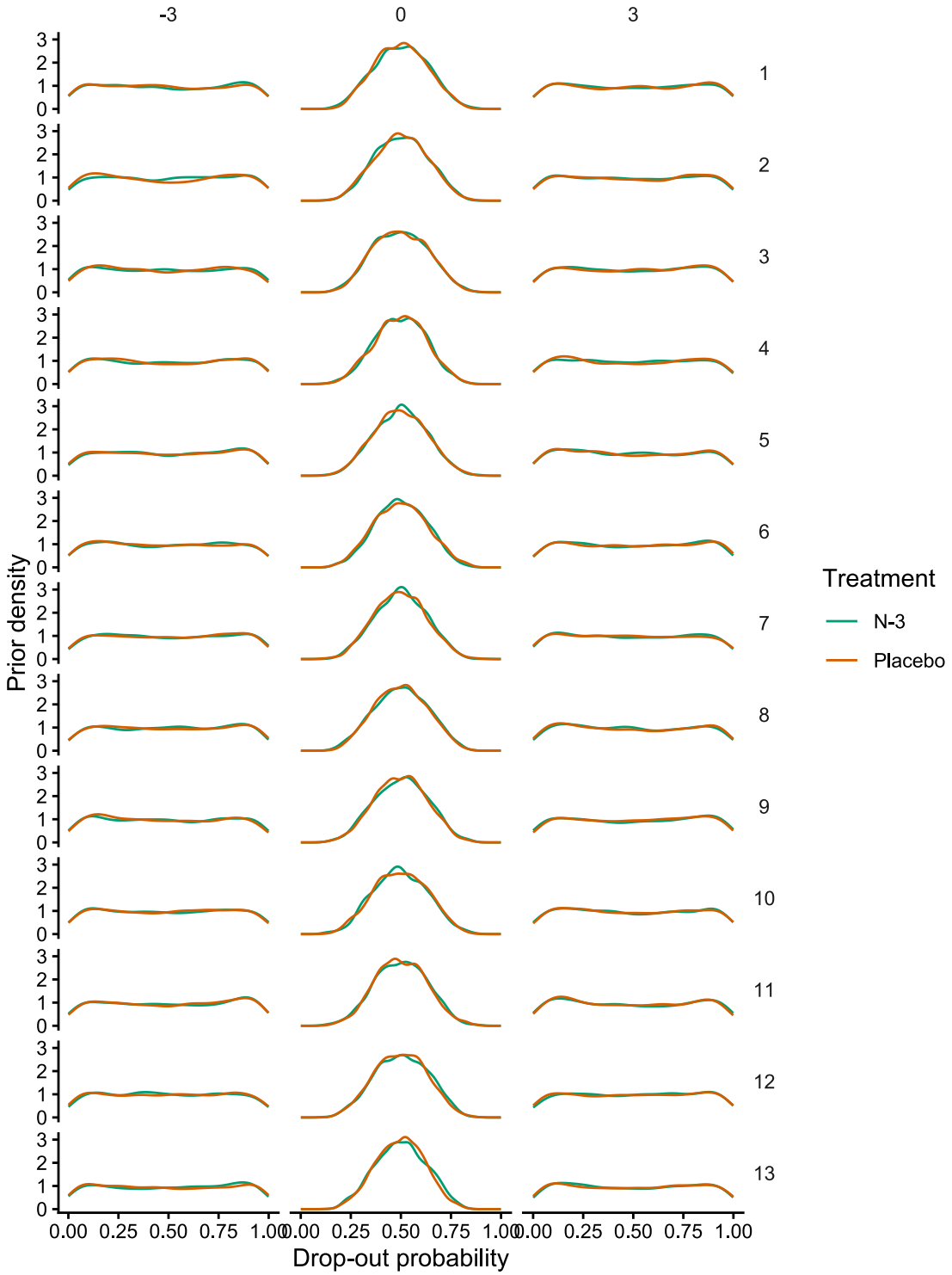


Figure 4.5: Prior predictive simulation for model 1. Columns are values of the predictor (SD-units) rows represents each predictor.

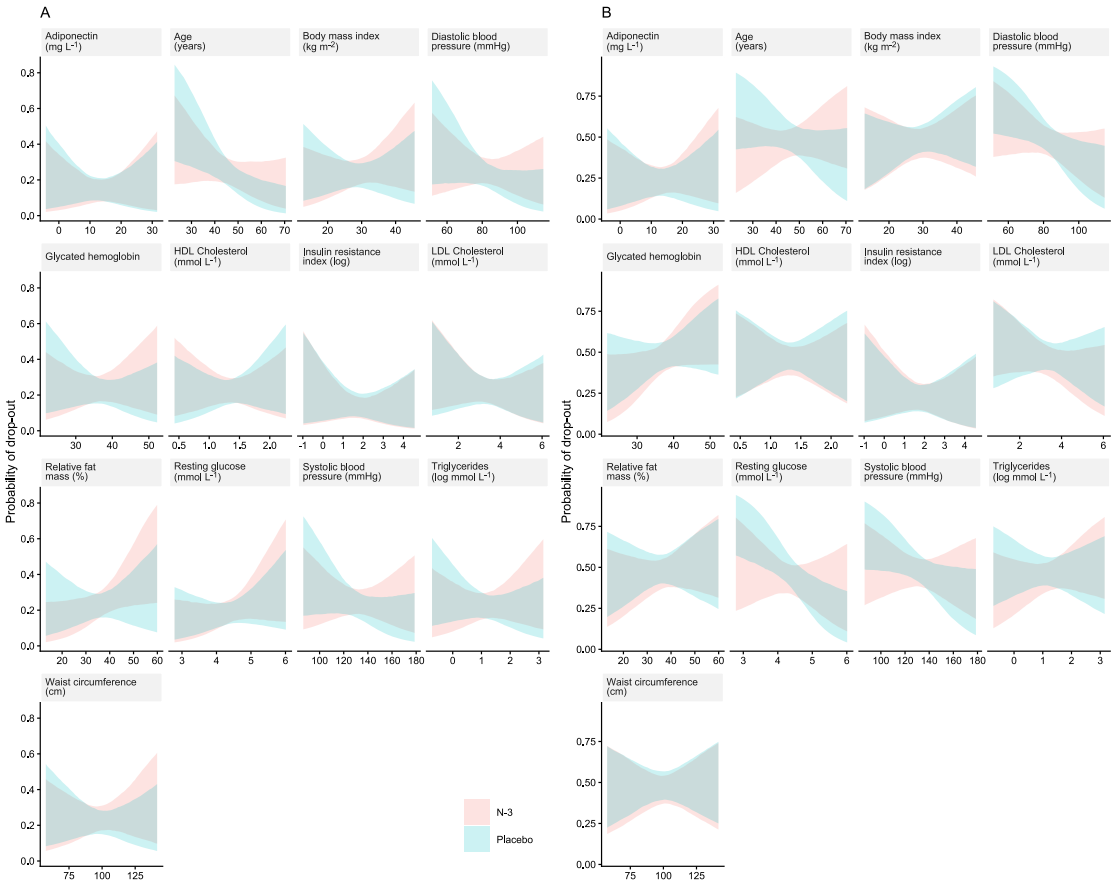


Figure 4.6: Probability of drop-out across values of baseline co-variates on per treatment group to time-point 2 (A) and 4 (B). Estimates are adjusted for sex, intervals are 90% percentile interval.

Table 4.3: Change in log-odds (Slope) for drop-out for 1 SD-unit change in baseline characteristic (Outcome) and differences in slope between treatment groups (&Delta Treatment) at time T2 and time T4. Values are posterior means with 95% percentile intervals.

Outcome	T2		T4	
	Slope	Δ Treatment	Slope	Δ Treatment
Adiponectin (mg L-1)	−0.00 [−0.55, 0.49]	0.14 [−0.63, 0.94]	0.07 [−0.40, 0.54]	0.17 [−0.57, 0.91]
HDL Cholesterol (mmol L-1)	0.06 [−0.33, 0.43]	−0.16 [−0.81, 0.51]	−0.03 [−0.35, 0.31]	−0.04 [−0.61, 0.55]
Insulin resistance index (log)	−0.14 [−0.70, 0.38]	−0.00 [−0.78, 0.78]	−0.10 [−0.59, 0.34]	−0.05 [−0.75, 0.69]
Glycated hemoglobin	−0.02 [−0.41, 0.35]	0.22 [−0.41, 0.87]	0.28 [−0.07, 0.63]	0.19 [−0.39, 0.77]
LDL Cholesterol (mmol L-1)	−0.14 [−0.52, 0.25]	−0.04 [−0.72, 0.60]	−0.18 [−0.50, 0.17]	−0.11 [−0.67, 0.43]
Triglycerides (log mmol L-1)	−0.02 [−0.41, 0.35]	0.27 [−0.37, 0.94]	0.07 [−0.25, 0.38]	0.23 [−0.34, 0.76]
Age (years)	−0.41 [−0.76, −0.06]	0.31 [−0.29, 0.89]	−0.08 [−0.39, 0.25]	0.45 [−0.13, 1.05]
Body mass index (kg m-2)	0.08 [−0.29, 0.42]	0.20 [−0.42, 0.82]	0.10 [−0.21, 0.42]	−0.07 [−0.62, 0.46]
Diastolic blood pressure (mmHg)	−0.23 [−0.61, 0.15]	0.27 [−0.33, 0.84]	−0.35 [−0.69, −0.00]	0.23 [−0.34, 0.81]
Relative fat mass (%)	0.24 [−0.16, 0.64]	0.35 [−0.28, 1.00]	0.11 [−0.23, 0.48]	0.10 [−0.44, 0.67]
Fasting glucose (mmol L-1)	0.24 [−0.17, 0.65]	0.17 [−0.46, 0.77]	−0.34 [−0.74, 0.03]	0.42 [−0.18, 1.04]
Systolic blood pressure (mmHg)	−0.18 [−0.57, 0.20]	0.30 [−0.35, 0.92]	−0.23 [−0.58, 0.11]	0.30 [−0.25, 0.85]
Waist circumference (cm)	0.02 [−0.40, 0.40]	0.18 [−0.42, 0.83]	0.03 [−0.31, 0.35]	0.01 [−0.57, 0.59]

In conclusion, age was negatively associated with probability of drop-out at T2 without robust difference between treatment groups (Figure 4.6 A; Table 4.3). Resting glucose at baseline tended to predict drop-out to T4 with no strong difference between treatment groups (Figure 4.6 B; Table 4.3). Together this suggest that baseline covariates did not robustly interact with treatment in predicting drop-out. However, baseline covariates (age and fasting glucose) were associated with higher drop-out rates.

4.1.1 Synthetic data

Synthetic data is generated with known parameter values to validate models and to enable exploration of the effects of model misspecification and missing data. The first data simulation function uses a multivariate normal distribution to simulate conditional means for K measurement types and N participants.

$$\begin{aligned} y_{i,j} &\sim \text{MVNormal}\ (\mu_{k[i,j]}, \Sigma) \\ \mu_{k[i,j]} &= \alpha + \beta_k X \end{aligned} \tag{4.2}$$

Where $y_{i,j}$ is a vector of outcome values for participant i in time-point j, drawn from a multivariate normal distribution with conditional means ($\mu_{k[i,j]}$ and a variance-covariance matrix (Σ). The conditional means are simulated using a linear predictor with parameter vector β and design matrix X.

The covariance matrix is constructed from two separate correlation matrices and a cross-time-point vector of correlations within each measurement type. Residual standard deviations for each type (and time) are used to combine standard deviations and correlations into the variance-covariance matrix[1].

[1]In the simulation function (`./R/sim-fun/sim-type.R`) separate variance-covariance matrices are constructed and then combined. But separate correlation matrices are the function's input.

$$
\begin{aligned}
\rho_{t2} &= \begin{bmatrix} 1 & \rho_{12} & \cdots \\ \rho_{21} & 1 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \\
\rho_{t4} &= \begin{bmatrix} 1 & \rho_{12} & \cdots \\ \rho_{21} & 1 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \\
\rho_{\text{cross}} &= \text{diag } (p_{k,\text{ time}}) \\
R &= \begin{bmatrix} \rho_{t2} & \rho_{\text{cross}} \\ \rho_{\text{cross}}^T & \rho_{t4} \end{bmatrix} \\
S &= \text{diag } (\sigma_{k,\text{ time}}) \\
\Sigma &= S \times R \times S
\end{aligned}
\tag{4.3}
$$

We will be able to control the within-time *residual* correlations between measurement types and between-time *residual* correlations within measurement types. The full correlation matrix (R) will set the residual correlation between measurement types at different time points to 0. These will, however, be correlated through within-time-point and pre-intervention correlations.

The pre-intervention values are also conditionally constructed using a separate multivariate normal distribution.

$$
\begin{aligned}
y_{\text{pre }[i]} &\sim \text{MVNormal } (\mu_{k[i]}, \Sigma) \\
\mu_{k[i]} &= \alpha + \beta_k X \\
R &= \begin{bmatrix} 1 & \rho_{12} & \cdots \\ \rho_{21} & 1 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \\
S &= \text{diag } (\sigma_k) \\
\Sigma &= S \times R \times S
\end{aligned}
\tag{4.4}
$$

Using this definition, we can control the correlations among measurement types (at baseline) and their variation. We can also make the means conditional on a set of predictors (design matrix X) with corresponding parameter values (β_k).

The approach detailed above is convenient as we are able to define each pairwise correlation. However, this comes with two caveats. First, for 8 variables we need to set $((8 \times 8 - 8)/2) \times 3 = 84$ correlation coefficients to capture between variable correlations at each time-point. Second, this approach does not guarantee a positive definite covariance matrix. An alternative is to use a factor loading approach[2]. For each variable we can specify a loading onto a factor and a residual variance for each variable (type). This approach is determined by λ, a $K \times F$ matrix where k_i are variables and f_j are latent factors, and ψ, a vector of length K defining residual variation.

$$
\begin{aligned}
\Sigma_{\text{raw}} &= \lambda \times \lambda^T + \text{diag } (\psi) \\
D &= \sqrt{\text{diag } (\Sigma_{\text{raw}})} \\
R &= D^{-1} \times \Sigma_{\text{raw}} \times D^{-1} \\
S &= \text{diag } (\sigma_k) \\
\Sigma &= S \times R \times S
\end{aligned}
\tag{4.5}
$$

This approach is used in `sim_type2` where Σ is built from the factor approach. Lambda is a matrix with K rows, representing variable types and F columns representing factors.

```
build_sigma_factor <- function(Lambda, psi, S) {
  K <- nrow(Lambda)
  R_raw <- Lambda %*% t(Lambda) + diag(psi)
  d <- sqrt(diag(R_raw))
  R <- R_raw / outer(d, d)
```

[2]See here for an introduction on the maths.

```
  diag(S) %*% R %*% diag(S)
}
```

As an example, $K = 5$ variable types are defined using $F = 3$ factors below. Approximate "variance explained" can be used to define the relationship between variables and factors where for k_1 we aim for 50%, 20% and 10% explained by f_1 to f_3 and the remaining 20% is error. The squared loadings plus error sum to 1: $0.5 + 0.2 + 0.1 + 0.2 = 1$. In code:

```
# Loadings as proportions
l1 <- c(0.5,  0.2,  0.1)
# Transforms to loadings
l1 <- sqrt(l1)
error <- 0.2

sum(l1^2) + error
```

```
[1] 1
```

The above example first defines loadings as proportions (summing to 1 with errors) and then adds a step to transform the proportions to loadings. For a full $K \times F$ λ matrix we populate the matrix and use ψ for errors.

```
# Defining a matrix of proportions explained by factors
Lambda <- matrix(
  c(0.5, 0.2, 0.1,
    0.8, 0.1, 0.05,
    0.1, 0.15, 0.1,
    0.05, 0.05, 0.2,
    0.05, 0.1, 0.01
    ),
                ncol = 3,
                byrow = TRUE)
# Defining remaining error
psi <- c(0.2, 0.05, 0.65, 0.7, 0.84)

# Transform from proportions to loadings
Lambda <- sqrt(Lambda)

# Check total proportions
rowSums(Lambda^2) + psi
```

```
[1] 1 1 1 1 1
```

Each row of the λ matrix plus errors now sum to 1. Next, we can use the λ matrix to construct Σ with unit standard deviations. The full covariance matrix (Σ) is used in calling `MASS::mvnorm` to produce the multivariate normal variable. As we are using standard deviations equal to 1, the covariance matrix equals the correlation matrix.

```
S <- rep(1, 5)
Sigma <- build_sigma_factor(Lambda, psi, S)

out <- MASS::mvrnorm(1000, rep(0, 5), Sigma = Sigma)

colnames(out) <- paste0("k", 1:5)
```

Table 4.4: Correlations used to simulate data built using a factor approach.

	k1	k2	k3	k4	k5
k1	1.000	0.845	0.497	0.400	0.331
k2	0.845	1.000	0.476	0.371	0.322
k3	0.497	0.476	1.000	0.299	0.225
k4	0.400	0.371	0.299	1.000	0.165
k5	0.331	0.322	0.225	0.165	1.000

Table 4.5: Correlations from simulated data (n=1000) built using a factor approach.

term	k1	k2	k3	k4	k5
k1	NA	0.845	0.514	0.389	0.325
k2	0.845	NA	0.502	0.345	0.323
k3	0.514	0.502	NA	0.310	0.265
k4	0.389	0.345	0.310	NA	0.141
k5	0.325	0.323	0.265	0.141	NA

In the loadings we defined k_1 and k_2 to be highly correlated through being influenced to a large degree by factors f_1 and f_2. In contrast, k_5 was given a lot of variation not explained by the factors which resulted in lower correlations with the other variables.

4.1.1.1 Heteroscedasticity and non-normal noise

The synthetic data-generating models described above produces homoscedastic data, i.e., equal residual variation between treatment groups. To create synthetic data under heteroscedastic conditions (unequal variance between treatment groups) an additional data generating model is created to enable setting treatment-specific variation (`sim_type3` function in `R/sim-fun.R`). The `sim_type3` function also includes two arguments to set additional noise in post-treatment measurements `prop_noise` controls the fraction of measurements that gets extra noise added and `sd_noise` controls the zero-centered normal distribution from which additional noise is drawn.

4.1.1.2 Missing data

Two mechanisms for including missing values are used in the simulation functions. First, a proportion of pre- and post-intervention values are set to missing, affecting single observations (e.g., a single measurement within a single participant). Second, the probability of missing observations from a whole participant is simulated using a Bernoulli model. This mechanism can connect baseline predictors to missing values. A Bernoulli simulation is performed, with p_i modeled linearly by α, which controls the average probability of complete missing observations, and the additional parameters β linked to predictors in a predictor data frame.

$$\begin{aligned} y_{\text{miss }[i]} &\sim \text{Bernoulli }(p_i) \\ \text{logit }(p_i) &= \alpha + \beta X \end{aligned} \tag{4.6}$$

Simulating a full (K = 8) data set of synthetic data, with missing values, results in observed patterns similar to the observed data (Figure 4.4, Figure 4.7 and Figure 4.8).

```
# A tibble: 3,552 × 8
   participant  type group   sex   age time       y    ystd
         <int> <int> <int> <int> <dbl> <chr>  <dbl>   <dbl>
```

```
 1           1     1     1     1  1.23 T1    -1.46  -1.23
 2           1     1     1     1  1.23 T2    -0.537 -0.568
 3           1     1     1     1  1.23 T3    -0.364 -0.444
 4           1     2     1     1  1.23 T1     0.762  0.524
 5           1     2     1     1  1.23 T2     0.114 -0.0497
 6           1     2     1     1  1.23 T3     1.51   1.19
 7           1     3     1     1  1.23 T1    -1.39  -1.17
 8           1     3     1     1  1.23 T2    -0.647 -0.547
 9           1     3     1     1  1.23 T3    -1.03  -0.869
10           1     4     1     1  1.23 T1     0.400  0.178
# ℹ 3,542 more rows
```

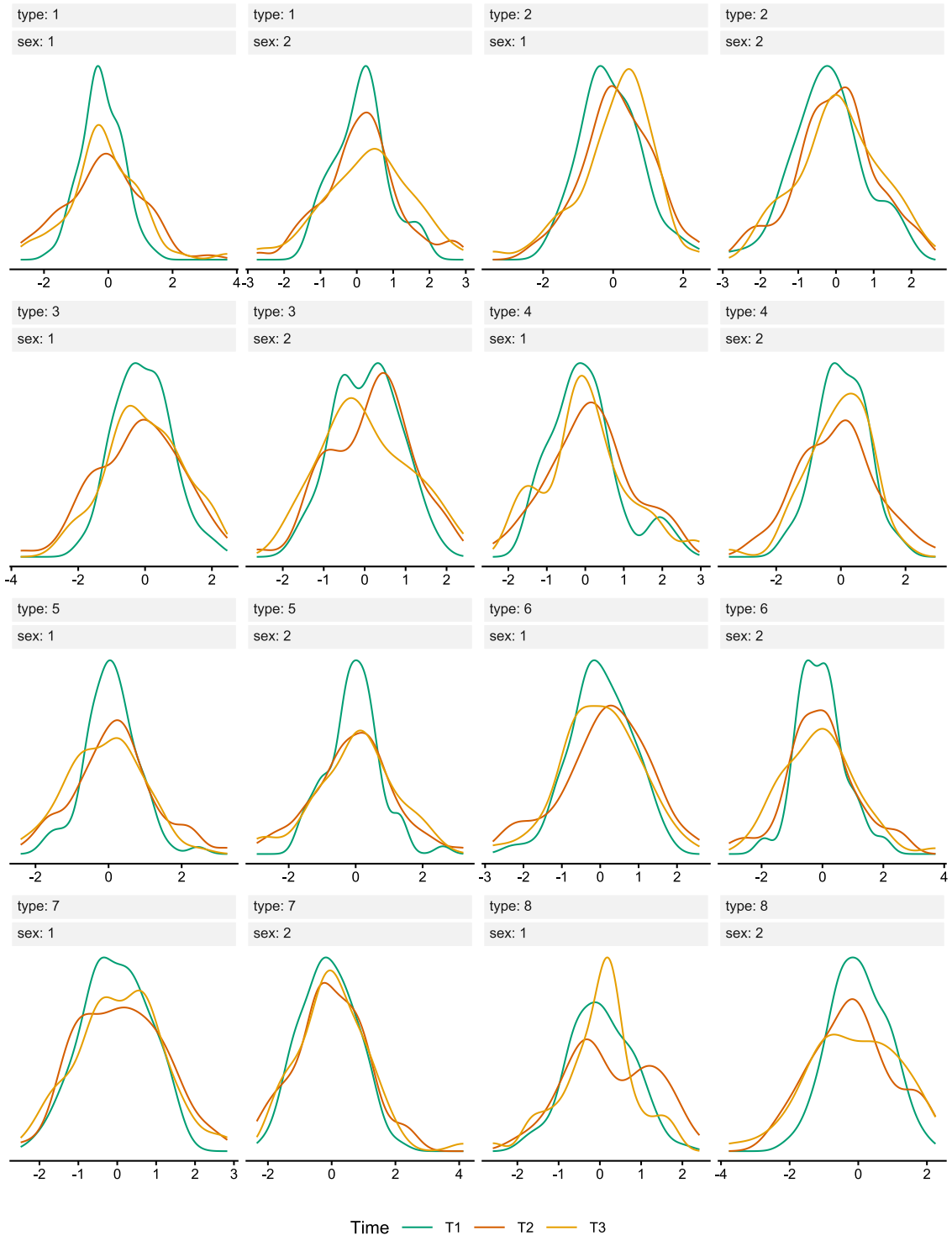


Figure 4.7: Distribution of (standardized) synthetic data across sex and measurement types.

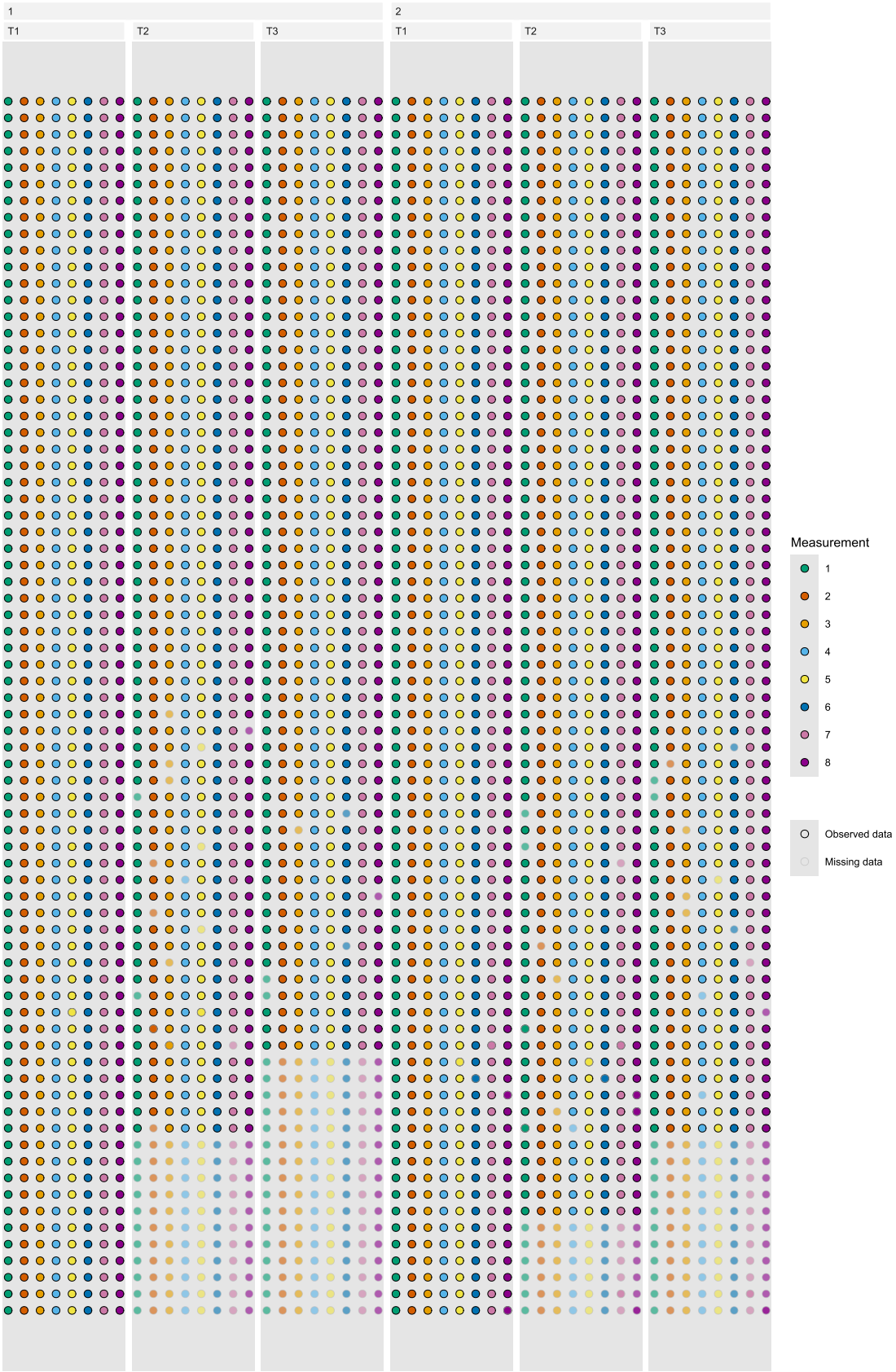


Figure 4.8: Missing data patterns from the data simulation. Missing values are found in specific measurements and in all measurements for a participant at specific time points.

5. Effects of N-3 supplementation on indications of metabolic syndrome - Models

Here models are incrementally developed by incorporating and testing features using prior predictive and posterior predictive checks, using both synthetic and observed data. Model performance (e.g., chain convergence and divergent transitions) is evaluated at relevant steps. The following chapter describes incremental model expansions towards a set of models that will be used to draw inference about the effect of krill oil supplementation on metabolic health.

The first model (model s1, `models/s1.stan`) is an ANCOVA-type model with pre-intervention values as covariates for estimating the post-intervention treatment- and measurement-type-specific means. The model is defined as:

$$
\begin{aligned}
y_i &\sim \text{Normal } (\mu_i, \sigma_i) \\
\mu_i &= \alpha_{\text{type,[TX]}} + \beta_{\text{type}} \times \text{pre}_i \\
\alpha_{\text{type,[TX]}} &\sim \text{Normal } (0, 1) \\
\beta_{\text{type}} &\sim \text{Normal } (1, 0.4) \\
\sigma_{\text{type}} &\sim \text{Exponential } (1)
\end{aligned}
\tag{5.1}
$$

Where $\alpha_{\text{type,TX}}$ is the measurement-type and treatment (TX) -specific intercept, and pre_i is the row-specific pre-treatment value.

Prior to modeling, observed values are placed on a standardized scale (mean-centered and scaled to the standard deviation). Since the model specification gives the same Normal $(0, 1)$ prior for both treatment groups, the implied prior for the differences between groups is Normal $(0, \sqrt{2})$. 95% of the prior probability mass therefore captures differences between groups from about −2.8 to 2.8. This prior is considered weakly regularizing. Similarly, the prior for the type-specific parameter for the pre-value covariates expects a 1-to-1 relationship between pre- and post-intervention values but with a wide range of possible relationships (95% prior probability mass in the range 0.2 to 1.8). The prior for the σ is also considered weakly regularizing allowing for a wide range of possible values but with more prior weight below the average (1).

The second model (model S2, `models/s2.stan`) for the main outcomes uses a multivariate normal likelihood. Here, we model outcomes as sampled from a multivariate normal distribution with a covariance matrix that connects the outcome types. This model extends model S1 by modelling the correlation structure among measurement types, enabling partial pooling across them. In further extensions of this model, the correlation structure will provide precision for imputing missing values, as observed scores will inform them through the structure. The same ANCOVA-type model is used here, in which pre-intervention scores serve as a covariate for the post-intervention group and measurement-type-level averages.

The model is defined as:

$$
\begin{aligned}
y_p &\sim \text{MVNormal}\ (\mu_p, \Sigma) \\
\mu_{p,\ \text{type}} &= \alpha_{\text{type,TX}} + \beta_{\text{type}} \times \text{pre}_{p,\ \text{type}} \\
\alpha_{\text{type,TX}} &\sim \text{Normal}\ (0, 1) \\
\beta_{\text{type}} &\sim \text{Normal}\ (1, 0.5) \\
\Sigma &= \text{S} \times \text{R} \times \text{S} \\
S &= \begin{bmatrix} \sigma_{\text{type}\ [1]} & 0 & .. \\ 0 & \sigma_{\text{type}\ [2]} & .. \\ .. & .. & .. \end{bmatrix} \\
R &= \begin{bmatrix} 1 & \rho_{12} & .. \\ \rho_{21} & 1 & .. \\ .. & .. & .. \end{bmatrix} \\
\sigma_{\text{type}} &\sim \text{Exponential}\ (1) \\
\text{R} &\sim \text{LKJcorr}\ (4)
\end{aligned}
\tag{5.2}
$$

Where y_p is a participant-level vector of outcome scores. Σ is the covariance matrix with standard deviations S and the correlation matrix R. The prior on R is defined using the LKJ distribution which provides regularization away from extreme values (-1, 1).

The third (model S3, `models/s3.stan`) includes all available data for one time-point, and imputes missing data. Imputation is done for both missing pre- and post-intervention observations. The pre-intervention imputation is done using a separate multivariate-normal model, defined as:

$$
\begin{aligned}
y^{\text{all}}_{\text{pre},p} &\sim \text{MVNormal}\ (\mu_{\text{pre},p}, \Sigma_{\text{pre}}) \\
y^{\text{all}}_{\text{pre},\ \text{type},p} &= \begin{cases} y^{\text{obs}}_{\text{pre},\ \text{type},p} \text{ if observed} \\ y^{\text{miss}}_{\text{pre},\ \text{type},p} \text{ if missing} \end{cases} \\
y^{\text{miss}}_{\text{pre},\ \text{type},p} &\sim \text{Normal}\ (0, 2)
\end{aligned}
\tag{5.3}
$$

Where missing values are treated as parameters and given a vague prior (Normal(0,2)), this makes sampling under prior-only more efficient.

The main model uses the $y^{\text{all}}_{\text{pre},p}$ as a covariate in the linear predictor. Again, missing values in the outcome are treated as parameters. The prior for missing observation is only included in prior predictive simulations.

$$
\begin{aligned}
y^{\text{all}}_{p} &\sim \text{MVNormal}\ (\mu_p, \Sigma) \\
y^{\text{all}}_{p,\ \text{type}} &= \begin{cases} y^{\text{obs}}_{p,\ \text{type}} \text{ if observed} \\ y^{\text{mis}}_{p,\ \text{type}} \text{ if missing} \end{cases} \\
y^{\text{miss}}_{\text{type},p} &\sim \text{Normal}\ (0, 2) \\
\text{pre}_{p,\ \text{type}} &= y^{\text{all}}_{\text{pre},p,\ \text{type}} \\
\mu_{p,\ \text{type}} &= \alpha_{\text{type,TX}} + \beta_{\text{type}} \times \text{pre}_{p,\ \text{type}} \\
\alpha_{\text{type,TX}} &\sim \text{Normal}\ (0, 1) \\
\beta_{\text{type}} &\sim \text{Normal}\ (1, 0.5) \\
\Sigma &= \text{S} \times \text{R} \times \text{S} \\
S &= \begin{bmatrix} \sigma_{\text{type}\ [1]} & 0 & .. \\ 0 & \sigma_{\text{type}\ [2]} & .. \\ .. & .. & .. \end{bmatrix} \\
R &= \begin{bmatrix} 1 & \rho_{12} & .. \\ \rho_{21} & 1 & .. \\ .. & .. & .. \end{bmatrix} \\
\sigma_{\text{type}} &\sim \text{Exponential}\ (1) \\
\text{R} &\sim \text{LKJcorr}\ (4)
\end{aligned}
\tag{5.4}
$$

Model S4 (`models/s4.stan`) adds sex as a covariate with the intention to improve predictions. The relevant addition to model s3 is in the linear predictor:

$$\begin{aligned} \mu_{p,\text{ type}} &= \alpha_{\text{type,TX}} + \beta_{\text{type}} \times \text{pre}_{p,\text{ type}} + \beta_{\text{sex}} \times \text{sex}_p \\ \beta_{\text{type,sex}} &\sim \text{Normal }(0, 0.5) \end{aligned} \tag{5.5}$$

The sex_p variable is created as male $= -0.5$ and female $= 0.5$ which gives that the prior for $\beta_{\text{type,sex}}$ expects that most of the difference between males and females in post-intervention scores is within ± 1 SD.

Model S5 (`models/s5.stan`) and S6 (`models/s6.stan`) further extends previous models by including both post-intervention time-points. In model S5 residuals are not correlated between time-points. Instead, separate covariance matrices are built for each time-point. The covariance matrices are therefore

$$\begin{aligned} \Sigma_{\text{time}} &= \text{S}_{\text{time}} \times \text{R}_{\text{time}} \times \text{S}_{\text{time}} \\ S_{\text{time}} &= \begin{bmatrix} \sigma_{1\text{ [time]}} & 0 & .. \\ 0 & \sigma_{2\text{ [time]}} & .. \\ .. & .. & .. \end{bmatrix} \\ R_{\text{time}} &= \begin{bmatrix} 1 & \rho_{12\text{ [time]}} & .. \\ \rho_{21\text{ [time]}} & 1 & .. \\ .. & .. & .. \end{bmatrix} \\ \sigma_{\text{type [time]}} &\sim \text{Exponential }(1) \\ \text{R}_{\text{time}} &\sim \text{LKJcorr }(4) \end{aligned} \tag{5.6}$$

The total number of parameters in each Σ is 36 for 8 measurement types.

Model 6 includes a full covariance matrix for the multivariate normal model. The total number of parameters in the matrix is 136 for 8 measurements with σ indexed 1 to 16. For S:

$$S = \begin{bmatrix} \sigma_1 & 0 & \cdots \\ 0 & \cdots & \cdots \\ \cdots & \cdots & \sigma_{2K} \end{bmatrix} \tag{5.7}$$

Finally, model S7 (`models/s7.stan`) extends model S1 by including both post-intervention time-points and missing data imputation. No correlation structures are included in the model and parameters (σ, β α are time- and type-specific).

5.0.1 Prior predictive checks, preliminary models

Prior predictive checks are performed for each model to confirm expected relationships between the pre-intervention covariate (Figure 5.1 and Figure 5.2), and prior expectations of treatment means (Figure 5.3).

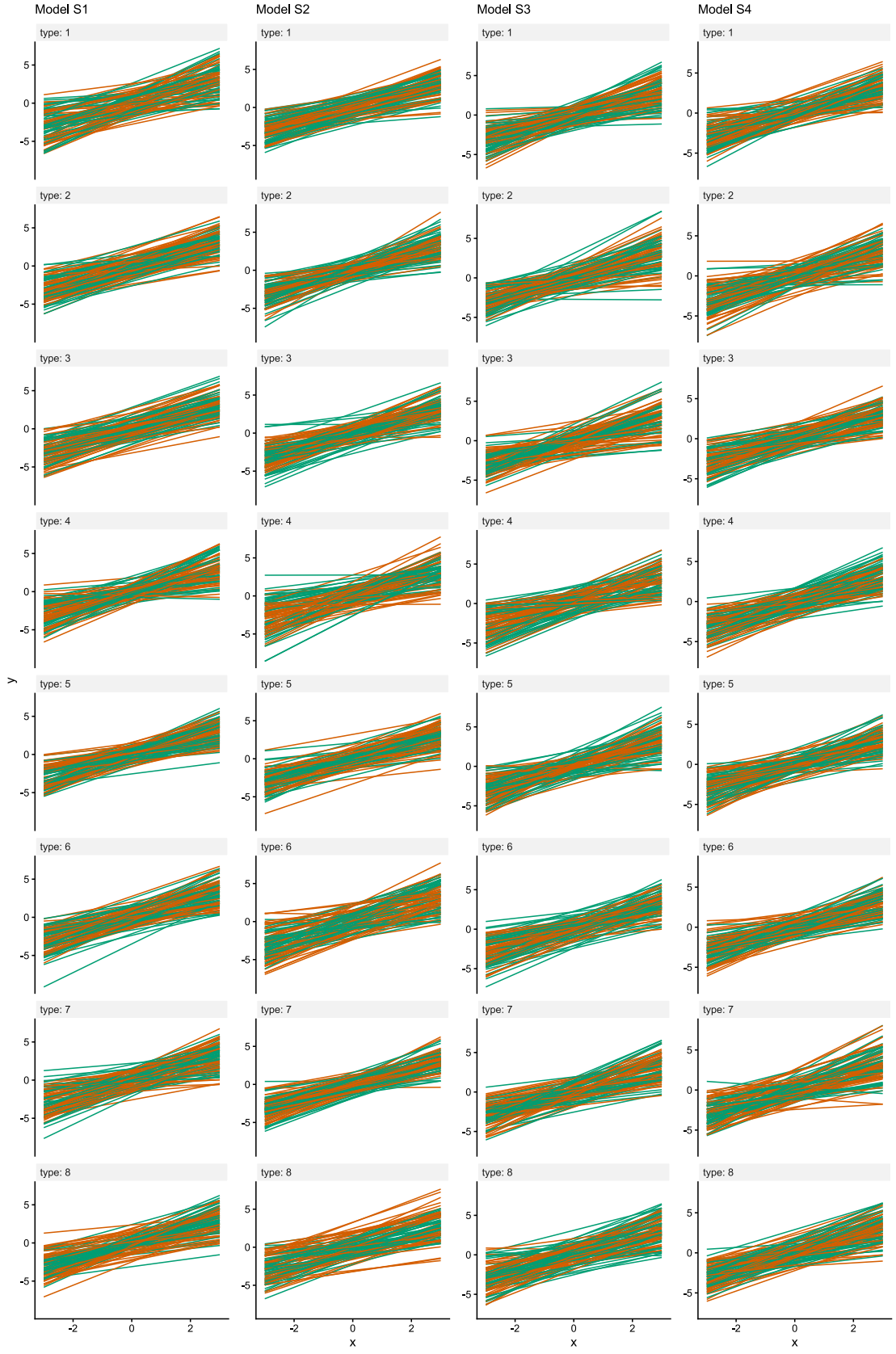


Figure 5.1: Prior predictive check of pre- to post-intervention associations in model S1-S4.



Figure 5.2: Prior predictive check of pre- to post-intervention associations in model S5-S7.

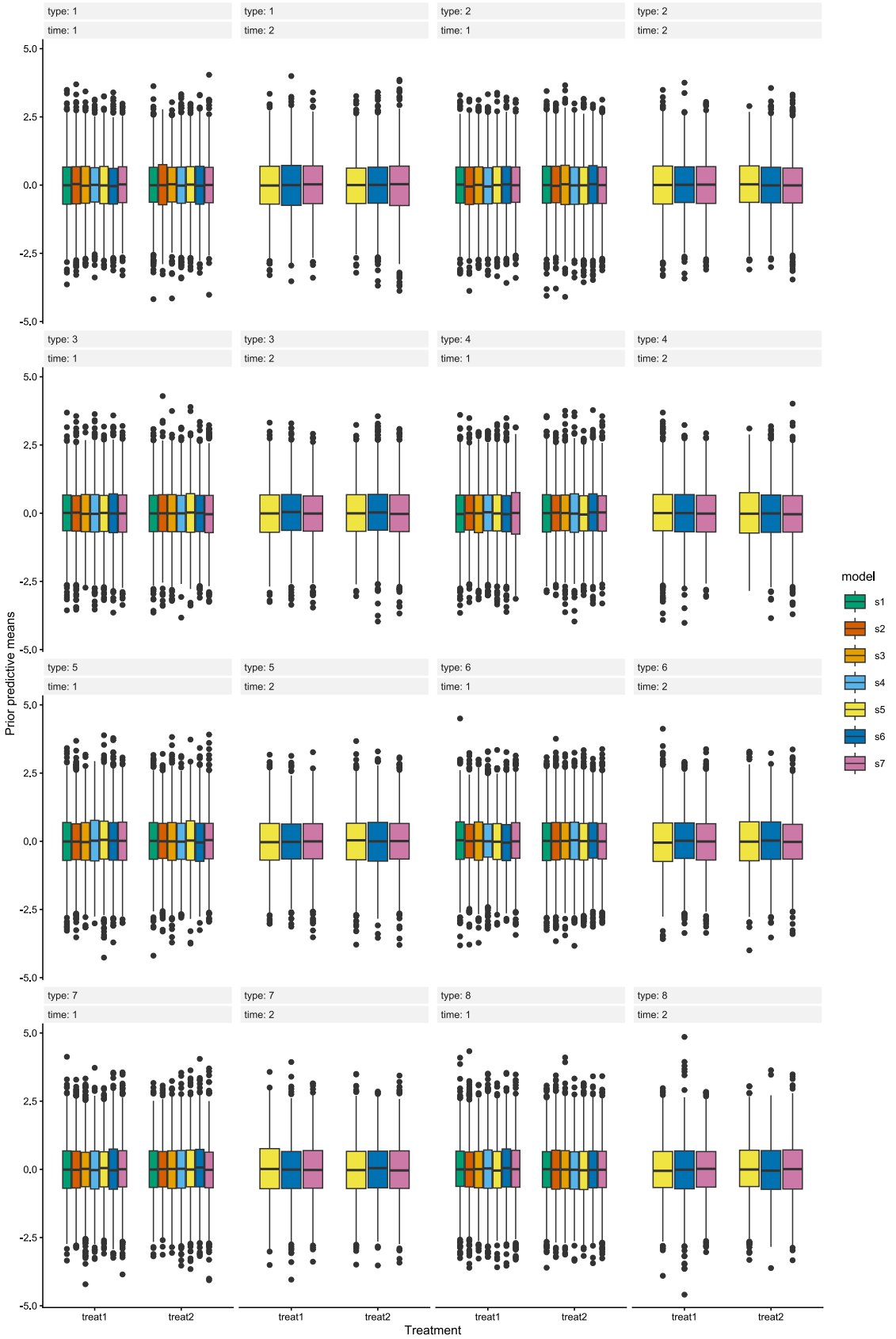


Figure 5.3: Prior predictive distributions of treatment means in models S1-S7.

The prior predictive checks confirm working and comparable models. A further check of model performance is done using synthetic data. We assess if we can recover simulated effects in the model estimates, we also assess the models effects of the prior on "reversed" effects, i.e. effect of opposite direction in the two groups.

Here, a simulated data set is constructed where we fix the relationship between pre- and post-intervention values ($\beta_{pre} = 0.95$), residual variation is set to 0.2 and effect sizes of treatment effects are set to −0.5 in type 1 and 7, 0.5 in type 2 and 8, and 0 in all other types. The model capture these effects and this preliminary check does not suggest preference for one treatment over the other (Figure 5.4).

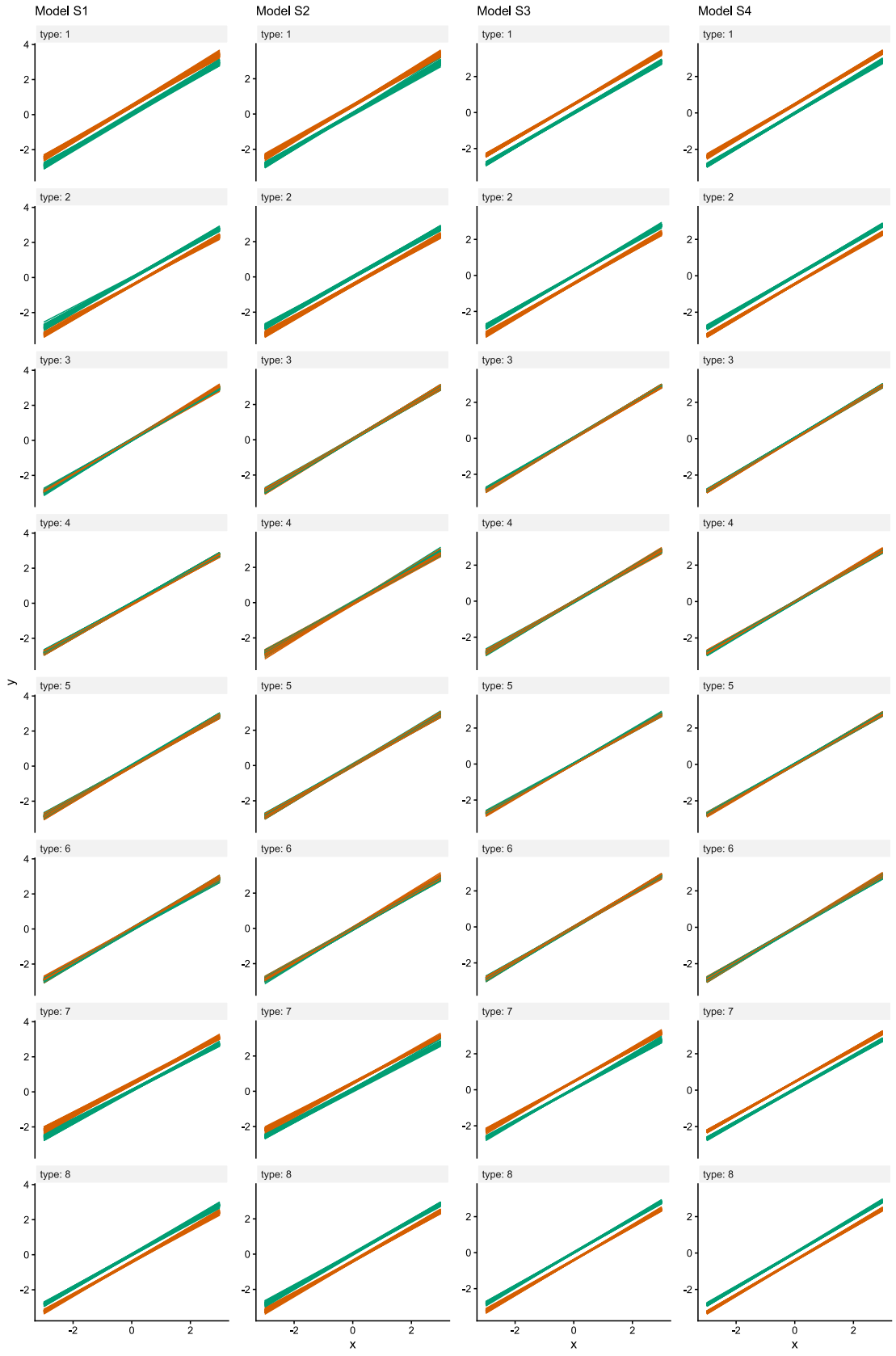


Figure 5.4: Posterior predictive check of pre- to post-intervention associations in model S1-S4 on synthetic data.

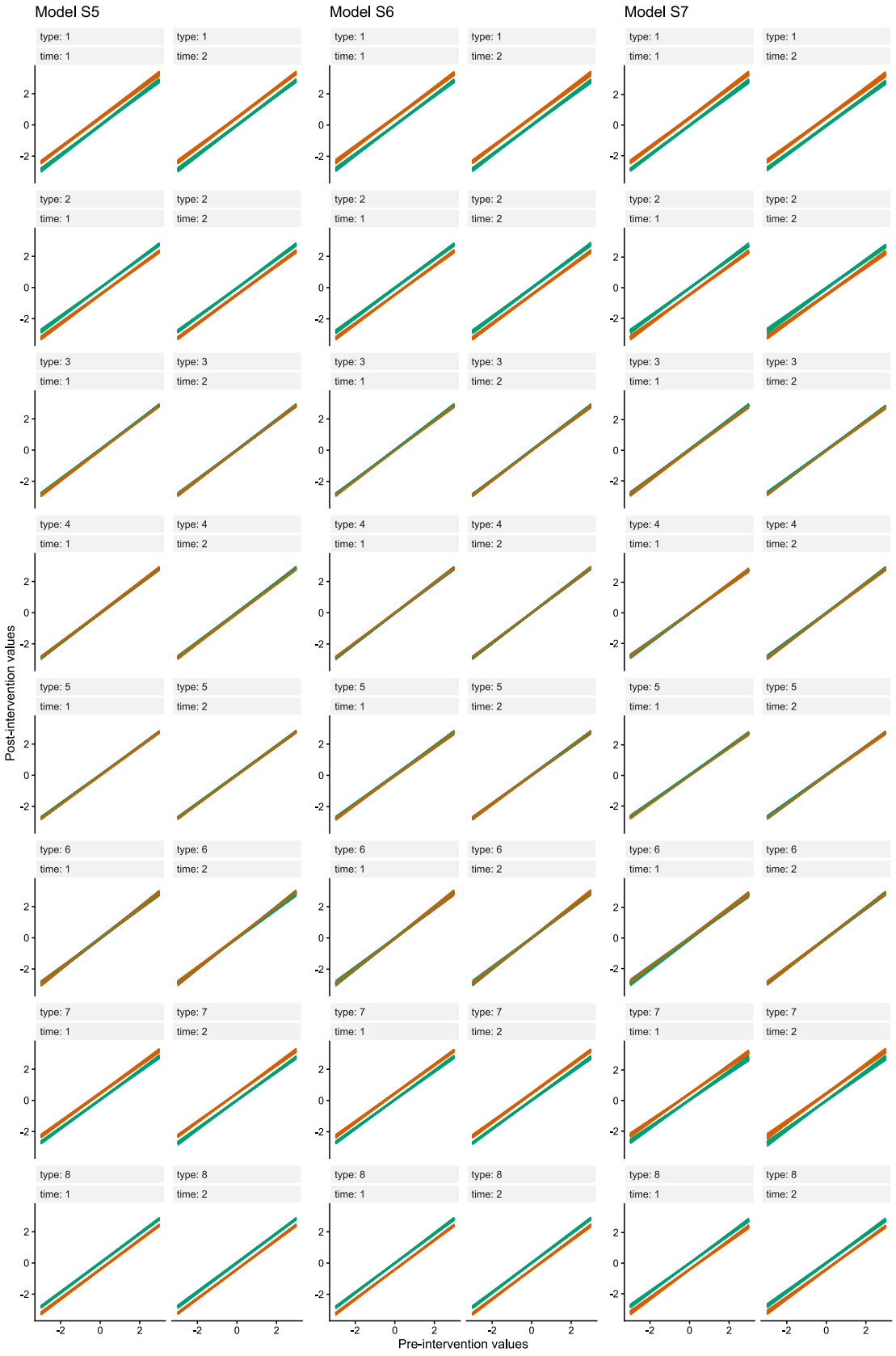


Figure 5.5: Prior predictive check of pre- to post-intervention associations in model S5-S7.

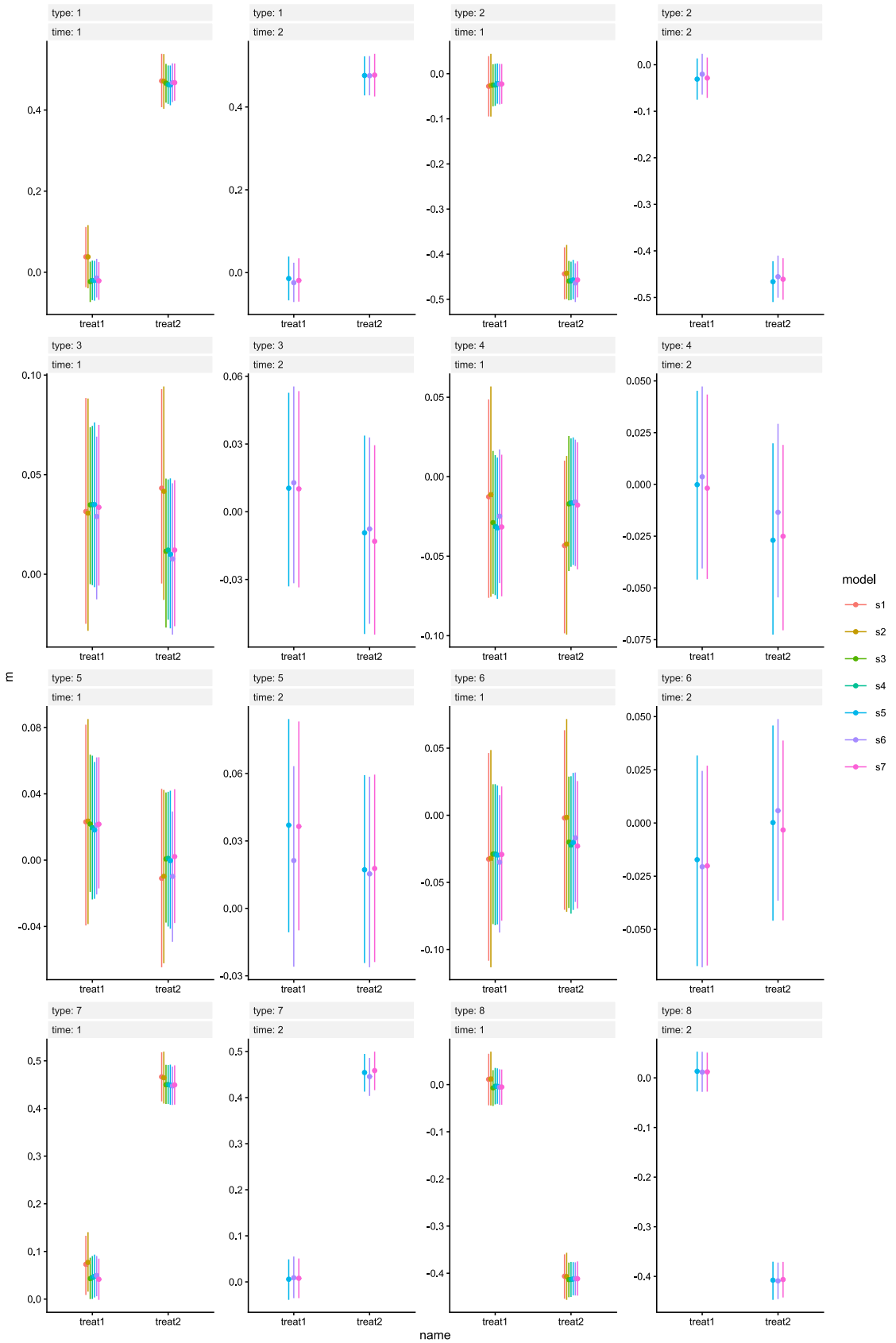


Figure 5.6: Posterior predictive distributions of treatment means in models S1-S7.

Using the synthetic data we confirm that the incrementally developed models captures the same patterns when they use the same data. In the above test, synthetic data was generated with large effects and negligible correlation structures across measurement types. We should therefore not expect large differences between models that could potentially capture different between-measurement type correlations patterns.

Based on these preliminary models, model S6 and S7 are kept for further development and evaluation. Model S6 is replaced by S8 that will also generate diagnostic criteria thresholds. Additionally, to allow for treatment-specific variation we including treatment-specific σ by updating model S7 and S8 to S9 and S10, respectively. The relevant update in S7 to S9 is an type-, time- and treatment-specific σ ($\sigma_{\text{type, time, treat}} \sim \text{Exponential}(1)$. Similarly, model s10 adds type-, time- and treatment-specific σ to the model by incorporating it in the multivariate-normal model.

Bibliography

1. Carpenter B, Gelman A, Hoffman MD, Lee D, Goodrich B, Betancourt M, et al. Stan: A Probabilistic Programming Language. Journal of Statistical Software [Internet]. 2017 ;76(1):1–32. Available from: https://www.jstatsoft.org/index.php/jss/article/view/v076i01
2. Gabry J, Cesnovar R, Johnson A, Bronder S. cmdstanr: R Interface to 'CmdStan' [Internet]. 2025. Available from: https://mc-stan.org/cmdstanr/
3. R Core Team. R: A Language and Environment for Statistical Computing [Internet]. Vienna, Austria; 2025. Available from: https://www.r-project.org/
4. Gelman A, Vehtari A, McElreath R, Simpson D, Margossian CC, Yao Y, et al. Bayesian Workflow. First edition. Boca Raton, FL ; Abingdon, Oxon: CRC Press; 2026.
5. Lemoine NP. Moving beyond Noninformative Priors: Why and How to Choose Weakly Informative Priors in Bayesian Analyses. Oikos. 2019 July ;128(7):912–28.
6. McElreath R. Statistical Rethinking: A Bayesian Course with Examples in R and Stan. Second edition. Boca Raton: CRC Press; 2020. (Chapman & Hall/CRC Texts in Statistical Science Series).
7. Gelman A, Carlin JB, Stern HS, Dunson DB, Vehtari A, Rubin DB. Bayesian Data Analysis. Third edition. Boca Raton: CRC Press; 2014.
8. Harris WS, Von Schacky C. The Omega-3 Index: A New Risk Factor for Death from Coronary Heart Disease?. Preventive Medicine. 2004 July ;39(1):212–20.
9. Alberti KGMM, Zimmet P, Shaw J. Metabolic Syndrome—a New World-wide Definition. A Consensus Statement from the International Diabetes Federation. Diabetic Medicine. 2006 May ;23(5):469–80.